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Constructing a Methodological Discourse for Using Emotion Recognition in AI Ethics—From Behaviorist to Cognitive Approaches

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ABSTRACT

Moral sentimentalism is a crucial field for building moral competence into artificial intelligence (AI) artefacts through emotion recognition and modeling techniques. Although the relation between emotions and the construction of *morally relevant thoughts* is not straightforward, various business applications are implied to be capable of detecting morally relevant mental content of their users through automated emotion recognition (AER). These applications rely on problematic ideas; while they involve effective technical solutions (e.g., image recognition, natural language processing), the foundations of their operationalization processes (i.e., selection of data collection and analytical methods for connecting emotions and moral thinking) are invalid. Following calls for a multidisciplinary methodological discourse on affective computing and emotionally laden human-technology interaction design, we propose a discussion of the use of automated emotion recognition in AI ethics. We start the discussion by considering two questions: (1) *Are the attempts to design moral competence in AI artefacts by means of automated emotion recognition techniques relevant?* (2) *If so, what kinds of limits should be acknowledged?* By clarifying the methodological landscape, we reveal that, *at best, automated emotion recognition can provide only hints about a person's emotional information processing*. Thus, the most urgent question to address is, *what is the nature of these hints and their connection to the person's moral reasoning and actions?* We coin the design concept of *inherent areas of reasoning* to guide proper operationalization processes for both technically and scientifically valid treatment of the question. Additionally, we elaborate how inherent areas of reasoning is related to another pivotal design concept: *emotion task*. A major threat is that an increasing amount of scientifically invalid applications will be adopted. The worry is urgent given that scientifically biased applications of emotion recognition systems have grown into large businesses.

KEYWORDS

Automated emotion inference (AEI); emotional artificial intelligence (EAI); emotions in HCI; AI ethics; affective computing

1. Introduction

A growing body of research literature has suggested using automated emotion recognition (AER)[1] techniques to build moral competence[2] into artificial intelligence (AI) artefacts. Underlying this phenomenon is an increasing acknowledgement of moral sentimentalism, according to which emotions play a crucial role in the rise of moral thinking, its contents, and morally relevant responses within humans (Kauppinen, 2022). While this reasoning may sound intuitive and easy to understand, the relation between emotional information processing[3] and the construction of *morally relevant thoughts* within an individual is not straightforward (Haidt, 2008; Liu et al., 2022; Monroe & Malle, 2019; Wallach et al., 2010). Still, this complexity has not slowed down the implementation of core ideas of moral sentimentalism into business applications, which are implied to be capable of detecting morally relevant mental content of their users through automated emotion recognition (Ajunwa, 2022; Boyd & Andalibi, 2023; Gloor, 2022; Roemmich et al., 2023; Ruckenstein & Velkova, 2019). These suggestions seem to be founded on mature technologies, such as facial recognition or natural language processing systems that can detect the emotional states of people. In reality, the technical artefacts

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are mature, but the operationalization processes of core concepts related to emotional information processing and moral sentimentalism underlying the data selection and collection, analytical methods, and evaluation of results, are not. This aspect leads to gross overestimations of what can be validly argued about people's thinking and experience, or what beliefs about people's value systems can be justified with the help of the technological solutions.

Following calls for multidisciplinary methodological discourse on affective computing and emotionally laden human-technology interaction design (Barrett et al., 2019; Boehner et al., 2007; Branford, 2024; Gratch et al., 2024; Jeon, 2017; Mohammad, 2022; Rousi et al., 2020; Stark, 2023), we propose a methodological discussion of the use of automated emotion recognition as part of practical solutions within AI ethics.[4] The discussion ultimately addresses two questions: (1) *Are the attempts to design moral competence in AI artefacts by means of automated emotion recognition techniques relevant?* (2) *If so, what kinds of limits should be acknowledged?* We further divide the question of relevancy to consider *scientific relevancy* and the perceived *desirability* of the proposed technologies. The suggested discussion arises from the concern that scientifically biased applications will increasingly claim to be capable of detecting morally relevant mental content of individuals by merely detecting their physiological responses. This concern is urgent given that invalid applications of automated emotion recognition systems have developed into large businesses that influence decision-making procedures within societal institutions (Ajunwa, 2022; Barrett et al., 2019; Katirai, 2024; Markets and Markets, 2023; McStay, 2020; Roemmich et al., 2023; Stark & Hoey, 2021; Stark & Hutson, 2022).

To determine whether the use of automated emotion recognition to develop moral competence to AI artefacts is scientifically relevant, we first consider the sensibleness of automated emotion recognition in general by engaging with and synthetizing relevant parts of the rich discussion of the scientific validity of automated emotion recognition (Barrett et al., 2019; Boehner et al., 2007; Branford, 2024; Cowen et al., 2019; Heaven, 2020; Keltner et al., 2023; Mohammad, 2022; Stark, 2023). We then observe the possible connections between a person's emotional information processing and their morally relevant mental representations, the contents of those representations, and their resulting actions. We discuss the possible connections from the perspective of moral sentimentalism, which centers on related questions (Haidt, 2008; Hindriks & Sauer, 2020; Kauppinen, 2022; Malle, 2021). Finally, we observe possibilities for connecting findings from moral sentimentalism to automated emotion recognition. For this, we provide an overview of contemporary suggestions for the use of automated emotion recognition in AI ethics (Balkenius et al., 2016; Coeckelbergh, 2010; Crnkovic & Çürüklü, 2012; Gloor, 2022; Gomila & Amengual, 2009; Holst, 2021; Liu, 2021; Maldonato & Valerio, 2018; Nallur & Finlay, 2023; Saariluoma, 2021; Tzeng, 2022; Wallach et al., 2010).

Regarding scientific relevance, we note that any application of automated emotion recognition must emphasize an ecological validity viewpoint since these technologies are mostly considered for use in normal life settings (Heaven, 2020). Accordingly, in our inquiries related to scientific relevance, we explicate prevailing presumptions within related fields and discuss emerging foundational issues[5] in terms of content-based psychology (Myllylä & Saariluoma, 2022; Saariluoma, 1997), appraisal theories of emotions (Jokinen & Silvennoinen, 2020; Moors et al., 2013; Scherer, 2009; Scherer & Moors, 2019), and hybrid approaches of moral sentimentalism (Kauppinen, 2022), which describe human thinking and action as an integration of cognitive-affective processes (Holtzman, 2018; Malle et al., 2021; Monroe & Malle, 2019; Zhu & Thagard, 2002) and frame thinking as a dynamic process. Both aspects have greater ecological validity than the current tradition of considering single events in the formation of moral dimensions of thinking (Boehner et al., 2007; Damasio, 2006; Liu et al., 2022; Mischel & Shoda, 1995; Power & Dalgleish, 2016; Scherer, 2009; Scherer & Moors, 2019; Schwartz, 2012).

Our main finding regarding the sensibleness of automated emotion recognition in general is that most contemporary approaches are scientifically biased. Prevailing invalid presumptions cause developers and deployers of the artefacts to mistake epistemological commitments of emotion theories (e.g., basic emotion theories that we *can gain meaningful information* about a limited set of emotions by observing people's expressions) for ontological ones (e.g., the observed expressions *are* a straight causal link to mental states of people, which implies that we can hold expressions as emotions). The same invalid presumptions are present in concrete application suggestions for the use of automated emotion recognition to build moral competence in AI artefacts. Furthermore, our discussion reveals that

overcoming Hume's guillotine (the is–ought problem) is foundational to the operationalization of moral sentimentalism in general (Pigden, 2010).

In addition to our analysis and synthesis of contemporary challenges, we recommend correction measures for the field of automated emotion recognition and its application for building moral competence in AI artefacts. For instance, considerations of the emotional information processing of people should adopt an *action* point of view instead of a *behavior* one and thus replace behaviorist approaches of emotions with cognitive ones. A behaviorist approach views people as passive beings who react to external stimuli, whereas an action approach considers emotional information processing as part of people's active engagement with their environment (Boehner et al., 2005, 2007; Graham, 2023; Miller, 2003; Zhu & Thagard, 2002). Moreover, understanding the role of mental representations and its mechanisms in people's actions makes it possible to explain how the mind can control and guide human actions (Bandura, 1991; Saariluoma, 1997; Wallach et al., 2010; Zhu & Thagard, 2002). People not only represent cognitive content in their minds but also integrate the emotional dimensions of actions, events, and objects into the information content of mental representations. Thus, a way to circumvent Hume's guillotine is to combine emotions with actions to uncover the role of emotional valences[6] in constructing people's moral values (Saariluoma, 2021; Zhu & Thagard, 2002).

Research has shown that sentiment analysis can support inferences about non-personalized[7] valence attributions in discourses (Mohammad, 2022; Zhang et al., 2018). Additionally, recent experiments relying on appraisal theories of emotions have provided preliminary evidence of the ability to make inferences on results of single appraisal criteria underlying the experience of valence by observing people's facial micro-expressions (Aue et al., 2007; Aue & Scherer, 2008, 2011; Gentsch et al., 2015, 2020; Scherer et al., 2017; Scherer & Moors, 2019; Wu et al., 2012). However, even if these findings could be integrated as part of automated emotion recognition, they represent only a small part of a person's emotional information processing. There are always multiple forms of latent information processing involved in the encoding (i.e., elicitation and emergence) of emotions, particularly in relation to the eliciting event and the latent social context involved, and even in the seemingly straightforward phenomenon of natural emotion recognition (i.e., how people infer about the emotions of others) (Chen & Whitney, 2019; Folz et al., 2022; Hoegen et al., 2023; Mobbs et al., 2006; Mumenthaler & Sander, 2015; Scherer, 2009; Scherer et al., 2018). Thus, even in the best-case scenario, automated emotion recognition may only provide hints about a person's emotional information processing.

From these findings, we conclude that, with heavy changes in focus and goals,[8] attempts to incorporate moral competence into AI artefacts by means of automated emotion recognition are, scientifically speaking, relevant. To answer our second question regarding what kinds of limits should be acknowledged, we address the importance of the concept of *emotion task*[9] for any operationalization process related to the design or deployment activity of automated emotion recognition. We further show how forming an emotion task implicitly requires the framing of *inherent areas of reasoning* involved in designing and deploying automated emotion recognition in general. In Chapter 4 (Elaboration of limitations), we synthesize findings from earlier parts of the text to elaborate on five inherent areas of reasoning for operationalizing automated emotion recognition and moral sentimentalism in AI ethics: (1) *design of the emotion task*, (2) *inference on emotions and morally relevant mental content*, (3) *technical requirements*, (4) *ethical issues*, and (5) *appropriate development and use framed by the combination of limitations*. Our main finding regarding the limitations of using automated emotion recognition to build moral competence in AI artefacts is that, within operationalization processes aiming to develop concrete instantiations, the limits should be acknowledged according to the concrete emotion task and the combination of limitations set by the inherent areas of reasoning.

Such framing reflects the difficulties of developing more sophisticated technologies. Good intentions of designers can easily go astray during the operationalization of concepts and phenomena, as it becomes increasingly difficult to recognize the limits of one's own expertise. Thus, the operationalization phase of central concepts within AI design and development should be understood as requiring similar scientific rigidness as the development of scientific theories that inform those concepts (Michell, 1997; Stark, 2023).[10] This understanding demands a change in our perceptions of expertise and multidisciplinary collaboration within technology design and development (Eitel-Porter, 2021; Salo-Pöntinen

& Saariluoma, 2022). *Lending* concepts from other fields is not sufficient since scientific expertise is central to the *valid use* of those concepts (Birhane, 2021; Gitelman, 2013; Kaplan, 1964; Ketokivi, 2015; Michell, 1997; Neal et al., 2019; Saariluoma, 1997; Stark, 2023).

Altogether, our elaboration of the methodological landscape demonstrates that, *even in the best case-scenario, automated emotion recognition can provide only hints about a person's emotional information processing*. Thus, the most urgent question related to using automated emotion recognition as a practical solution in AI ethics is, *what is the nature of these hints and their connection to a person's moral reasoning and actions*? Additionally, we recognize *providing a flexible moral sensitivity*[11] *mechanism in AI artefacts* as a high-level emotion task for using automated emotion recognition in building moral competence for AI artefacts. As such, if the prevailing validity issues are overcome, automated emotion recognition is a valuable aspect for providing means to use AI in the process of constructing moral ideals (strong ethical AI), in addition to the prevailing idea of considering AI artefacts as the objects for readily formalized moral guidelines (weak ethical AI) (Saariluoma, 2021).

In the following section, we discuss our research approach for explicating the underlying presumptions in the operationalization processes of using automated emotion recognition in AI ethics. Furthermore, we present the concepts of emotional information processing, automated emotion recognition, and moral competence. In Section 1.2 we discuss how our research situates in relation to earlier human-computer interaction (HCI) literature, which is required, as the idea of using automated emotion recognition in AI ethics is contemporarily dealt with in scholarly traditions closely related to but not within HCI. We also explicate limitations of our approach and define how it can be complemented by more known approaches in HCI. Chapter 2 then attends to the validity discourse of automated emotion recognition initiatives before discussing pivotal design concepts and suggestions for corrections. In Chapter 3, we engage with moral sentimentalism and provide an illustrative overview of literature recommending the use of automated emotion recognition in AI ethics. Chapter 4 synthetizes our findings to frame inherently involved limitations to the development of moral competence in AI artefacts by means of automated emotion recognition. The article concludes by highlighting the central findings and perspectives for future research.

1.1. Research approach

With this study, we intend to explicate the major methodological issues underlying contemporary pursuits in designing and developing moral competence in AI artefacts by means of automated emotion recognition. This endeavor is crucial since it is widely recognized that a rigid *theoretical discussion* of this possibility must be multidisciplinary (Gloor, 2022; Nallur & Finlay, 2023). However, the *actual instantiation* of such ideas in concrete technologies is not understood to require similar multidisciplinary rigidness (Gloor, 2022; Liu et al., 2022). Parallel difficulties have been acknowledged in the field of affective computing in general (Barrett et al., 2019; Boehner et al., 2007; Branford, 2024; Gratch et al., 2024; Jeon, 2017; Mohammad, 2022; Monteith et al., 2022; Stark, 2023). Therefore, we discuss and explain the presumptions made in the operationalization processes of designing and developing moral competence in AI artefacts by means of automated emotion recognition. With the term *operationalization process*, we refer to the construction of empirical entities relating to the core theoretical concepts by defining relevant empirical attributes, selecting representative data and its collection and analytical methods, and forming a valid framework for evaluating the results (Hornbæk, 2006; Kaplan, 1964; Ketokivi, 2015; Saariluoma, 1997; Yin, 2009). This kind of investigation is called foundational analysis (Saariluoma, 1997).

In the present context, foundational analysis entails the explication and examination of the presumptions underlying the operationalization of central concepts for describing the use of automated emotion recognition to instantiate ideas of moral sentimentalism. A similar approach has been used by Sloane et al. (2022), for example, to investigate the operationalization of *job applicant(s)* in the context of automated hiring tools. Their study found the research method to be highly useful in determining that "assumptions that underpin a system … often rest on contradictory or pseudo-scientific theories about job applicants" and that these assumptions are not detected by the more-used technical audits (Sloane

et al., 2022, p. 1). As such, foundational analysis is a valuable approach for investigating the scientific validity of technological solutions.

Following Scherer (2009) and Scherer and Moors (2019), we use "emotional information processing" interchangeably with "affective information processing." Involuntary monitoring of emotionally relevant information from one's environment is constant (Rousi, 2013; Scherer, 2009; Zajonc, 2000). When the information processing exceeds the threshold of consciousness and produces an experience of feeling, it is referred to as an "emotional episode" (Oatley et al., 2006; Power & Dalgleish, 2016; Scherer & Moors, 2019). Like emotional information processing in general, emotional episodes are understood as dynamic, multi-component, multi-level appraisal processes that are elicited by individually relevant interactions that people have with their environment in an internal reality (i.e., memory) and/or external reality. The labeling of discrete emotions may or may not be part of emotional information processing; if it is, it is only one small part of it, and it occurs at the end of the appraisal process. The components of an emotional episode consist of the appraisal of the eliciting event, two aspects of action preparation (action tendency and physiological responses), expressive behavior, subjective feelings, and a possible verbal labeling of the experience with, for example, emotion words or references to dimensional aspects of the experience.

Emotion studies have commonly understood emotional episodes as information processing that reaches the threshold of forming conscious mental representations within an individual (Barrett et al., 2019; Cowen & Keltner, 2017; Oatley et al., 2006; Russell, 2003; Scherer & Moors, 2019). However, while the involuntary monitoring processes and their underlying mechanisms are always presumed, they are rarely fully explicated (Scherer, 2009). Attention involves continuously and involuntarily monitoring one's environment (Norman, 1968). At this stage, it is adequate to state that emotional information processing similarly involves continuously and involuntarily evaluating one's environment, though from a perspective of novelty, goal relevancy, and valence (Dan Glauser & Scherer, 2008; Rousi, 2013; Zajonc, 2000). In other words, people continuously evaluate affective cues from their environment.

With automated emotion recognition, we refer to technical artefacts that are built for the purpose of sensing and interpreting people's emotional information processing, which is sometimes combined with the term "detecting emotions". There is a wide variety of technical means for building such capabilities in technical artefacts, but a major distinction can be made between approaches that use textual information as the input and approaches that use overt physical indicators as the input for further interpretation (Mohammad, 2022; Picard, 1997). We address the employed methods in more detail in the next chapter. In this study, we choose to clearly distinguish between natural emotion recognition (i.e., how people infer about other people's emotions (De Prisco et al., 2023)) and automated emotion recognition (i.e., technical artefacts developed to make inferences about the emotional information processing of people). This terminological distinction has rarely been made in the research literature.[12] However, as we illustrate, the activities of natural emotion recognition and the development and use of automated emotion recognition have distinctions that make it theoretically and pragmatically reasonable to differentiate them at a symbolic level.

Moral competence is a conceptual framework in AI ethics that recognizes that while designing AI artefacts to have full moral agency is not feasible, there is a potential to design AI artefacts that can be sensibly considered morally capable in a narrow context and manner (i.e., to have some kind of moral competence) (Malle & Scheutz, 2019). Thus, it helps researchers and designers focus their work without disregarding the inherent limitations of their pursuits. Such concepts are highly valuable in multidisciplinary pursuits as they provide common ground for practical work and evaluation of results (Salo-Pöntinen, 2021).

1.2. Automated emotion recognition, moral sentimentalism, and HCI

The idea of using automated emotion recognition as a practical solution in AI ethics is contemporarily dealt with in scholarly traditions closely related to but not within HCI, as can be seen in our review provided in Section 3.2. Even so, the topic is most relevant to HCI[13] and one that HCI research traditions have much to offer. For example, HCI literature has produced rich critiques of prevailing presumptions in affective computing in the last two decades (Boehner et al., 2005, 2007; Gaver, 2009;

Höök et al., 2008) and as we describe in the next chapter, affective computing has a central role as a theoretical framework in contemporary automated emotion recognition. The critiques describe how the affective computing tradition has omitted inherent contingent characteristics of emotions while pursuing to describe emotions as suitable phenomena for computational engineering purposes. Computational engineering as a field accredits practical generalizations of subject phenomenon above all in its modus operandi (Cockton, 2004; Prabowo et al., 2023; Russell & Norvig, 2016), and affective computing has been considered to uncritically comply to this demand (Boehner et al., 2005, 2007; Harrison et al., 2007). In their responses, HCI perspectives have emphasized emotions to be formed in the *interactions* that people have with their environments, leading to considering emotions as *situated* and *related* in the prevailing cultural, social, and political conditions. Thus, emotions cannot be detached from said conditions (Boehner et al., 2005; Gaver, 2009; Höök, 2008; Höök et al., 2008).

Similar to existing critiques, our approach to emotions and moral aspects of thinking are heavily grounded on an interaction perspective and we discuss issues of situatedness and relatedness of emotions and moral thinking as part of our critique in Chapters 2 and 3. However, when the earlier literature in HCI has produced its critique with a focus on structure and politics and related to affective computing, we combine a review of relevant work in psychology and cognitive science and evaluate the use of automated emotion recognition within AI ethics. Notwithstanding our new focus area, the relevancy of situatedness and relatedness for our as well as earlier critiques illustrates a possibility to build bridges between cognitive scientific and social scientific HCI approaches on topics related to affective computing. We further elaborate on the necessity for interdisciplinary work on our topic by defining the approach of areas of reasoning in Section 2.1.1 and by providing an example of its use in Chapter 4. By doing so, we provide a set of orienting questions for new research that can be approached by reference to any research tradition in HCI.

Such bridging is needed, as on an epistemological stance, much of approaches to ethics in HCI are in favor of constructivism (Frauenberger et al., 2017; Shilton, 2018) and Critical Theories (Bardzell & Bardzell, 2011; Duarte & Baranauskas, 2016; Fiesler et al., 2016; Søndergaard, 2025; Sultana & Ahmed, 2019), which may seem to be contrary to what we discuss. However, epistemological stances in general should be considered to be situated on a spectrum rather than on a binary stance of in favor of/against and related to the subject of inquiry (which we believe the sited authors would agree on). For example, our approach is in line with presumptions of structuration theory as developed by Giddens (1984), which call to consider the ongoing balance of how social structures affect individuals and how individuals then again actively construe and maintain said social structures through their actions. This dynamic relationship is something that we discuss throughout Chapters 2 and 3 when dealing with the situatedness and relatedness of emotional information processing and moral thinking of individuals, and in Chapter 4 when discussing some of the implications of this dynamic interplay between structures and individuals for appropriate design and use of automated emotion recognition in AI ethics. However, our paper considers this dynamic relationship mostly from the point of view of the individual. Therefore, our approach is limited when considering the evaluation of social structures and politics in the development and use of automated emotion recognition technologies in AI ethics. Thus, our research should be complemented by views concentrated on the evaluation of the social sphere, coming from areas such as cultural anthropology, sociology, or feminism for example.

Placing moral sentimentalism to the more extensively known approaches to ethics in HCI requires acknowledging that HCI as a field is considered to contain multiple waves of research orientations (Bødker 2015; Gunkel, 2018), sometimes also referred to as waves of paradigms (Frauenberger et al., 2017; Harrison et al., 2007). On one hand, the waves are used to describe differences for what is considered as the central context of question setting in HCI. The first wave is often explained to consider the human-technology *fit*, with idea of fit being a feature to be designed into the artefact. The second wave, then again, has emphasized the need to understand the human mind and its workings as a central subject of interaction research and design. Research identified as belonging to the third wave, once again, largened the scope of HCI. Its advocates have emphasized that for interaction and its features to be truly understandable, they should be considered to develop over time in a given social and political environment (Bødker 2015; Gunkel, 2018; Harrison et al., 2007). This is not intended as *the description*

of HCI's development over history and we note that the number or shape of *waves* is not set in stone but depends on the interpretation of how approaches may and should be clustered (for competing clustering's cf. Bannon, 2011; Bødker 2015; Cockton, 2004; Harrison et al., 2007; Kuutti & Bannon, 2014; Saariluoma et al., 2016). However, we want to note that on the other hand, different waves of research orientations have been considered to frame sufficient and principal theoretical backgrounds for conducting proper HCI research related to given questions.

Questions related to ethics has often been considered as part of the 3rd wave HCI (Bødker 2015; Duarte & Baranauskas, 2016; Shilton, 2018). We pursue to show how moral sentimentalism opens the possibility of considering ethics as bridging orientations of the 2nd and 3rd waves to form complementary approaches. The bridging happens through the phenomena of situatedness and relatedness of emotions and moral thinking, which are considered central elements in the conceptual schema of meaning construction and mediation (Bach et al., 2024; Frauenberger et al., 2017; Fuchsberger & Frauenberger, 2025; Søndergaard, 2025; Vianello et al., 2023). Meanings are important, as they are considered to frame evaluative (questions of good and bad) as well as normative aspects (questions of right and wrong) of ethics and moral thinking. Thus, formation of and effects of meanings and structures that mediate them, are considered central aspects of ethics in HCI literature (Bardzell & Bardzell, 2011; Frauenberger et al., 2017; Fuchsberger & Frauenberger, 2025; Giaccardi et al., 2025; Gunkel, 2018; Sultana & Ahmed, 2019). Contemporarily, there are two central ontological orientations by which meaning is considered to manifest in the world: through *enacting* in the world as framed by grammars of action[14] which follows the footsteps of Wittgenstein's late work (Frauenberger et al., 2017; Fuchsberger & Frauenberger, 2025; Harper, 2019), and through *being* in the world (phenomenology), which emphasizes the experiential and relational domains of life (Giaccardi et al., 2025; Harrison et al., 2011; Søndergaard, 2025). Both stances and their fusions can be found in multiple forms in HCI research (feminism, ethnology, cultural anthropology, user experience, participative design, value-centered design etc.) (Bardzell & Bardzell, 2011; Bødker & Kyng, 2018; Cockton, 2004; Friedman & Hendry, 2019; Görnemann & Spiekermann, 2024; Hult et al., 2006).

We will show in detail in Chapters 2 and 3 that there are also traditions in cognitive science for considering the construction of and effects of meaning in the thinking of people. However, HCI literature that considers the philosophy of science in HCI often describes cognitive science as being dualistic in essence and favoring ideal formal models of humans in its research and approach to information processing, thus not being capable of dealing well with issues of meaning (Bardzell & Bardzell, 2011; Boehner et al., 2007; Cockton, 2004; Duarte & Baranauskas, 2016; Gunkel, 2018; Harrison et al., 2007, 2011). This statement holds true to connectionist and capacity-oriented branches of cognitive science (Saariluoma, 1995, 1997; Thagard, 2023), but similarly as HCI as a field, also cognitive science has developed during the past four decades in its scope for what is considered relevant research (Menary, 2010; Saariluoma, 1995, 1997; Zhu & Thagard, 2002). Simultaneous to the prevailing connectionist and capacity-oriented views on cognitive science, grew orientations that took the construction of and effects of meanings as a focal area of research and showed how neither the connectionist nor capacity-oriented branches can answer questions related to why certain information content is found in the thinking of people (Saariluoma, 1995, 1997). One such tradition, the content-oriented tradition, has explained that answers for information content must consider the relevancy of said content within its wider framework, i.e., the meaning of the content within the thinking of the person (Myllylä & Saariluoma, 2022; Saariluoma, 1997). Furthermore, the content-oriented tradition of cognitive science has a functional view on thinking, and as such, considering themes like meaning, emotions and the moral domain comes natural to it. We further explicate the nature and fruitfulness of this view of cognitive science in Section 2.1 and Chapter 3.

For now, it is enough to acknowledge that the connection of moral sentimentalism and the more known traditions of considering ethics in HCI can be explained to look at the rich phenomenon of ethics and human morality from different conceptual levels. The central entities within different levels cannot be reduced to entities on other levels. For example, social norms cannot be reduced to depict affective states of people even though there are connections between them. In other words, affective states can be said to be *necessary* but not *sufficient* for understanding dynamics of social norms.

However, as we will discuss in the following chapters, meaningful (nonreductive) connections between different conceptual levels can be drawn, that may enrich HCI discourses of AI ethics in contrary to the prevalent reductive stances within automated emotion recognition literature, which we critically evaluate in the next two chapters.

Thus, a well framed approach of moral sentimentalism is relevant for explaining underlying mechanisms that are at play when considering moral thinking on an individual level. Such a view is crucial for critical evaluation of contemporary approaches for using automated emotion recognition in AI ethics. Nevertheless, if considered as a sole explanatory approach, moral sentimentalism gives little consideration to the contents of ethical *discourse*, whereas methodological approaches known in sociology, anthropology, activity theory and critical studies are rich in this manner. For example, ethnography in its multiple forms is a method shared by many of the approaches relating to social sciences. It is used for understanding the dimensions of how values are considered in design practices (Friedman & Hendry, 2019; Shilton, 2013), or as giving insight into important values that should be catered for when designing for the examined group of people (Pink et al., 2013; Sultana & Ahmed, 2019). In addition, participatory design, and activity research (or practice research) emphasizes interactive design practices, where both, designers and user groups are considered active parties in developing visions of the valuable (Bødker & Kyng, 2018; Hult et al., 2006; Kuutti & Bannon, 2014). Once again, we emphasize that our approach should be considered as opening discussion that can be furthered and complemented by views stemming from other approaches known in HCI.

As a last note regarding the connection of our work to earlier research in HCI, the use of moral sentimentalism, or more precisely, considering the close connection between emotions and moral thinking as a relevant theoretical framework for research and design in HCI is not totally new. The work of Görnemann and Spiekermann (2024) is closest to our framing of moral sentimentalism which we describe in detail in Chapter 3. They state:" [W]e … argue that understanding the values that matter to an individual is a gateway to identifying the sources of their emotional reactions to technology", while putting an emphasis on how the connection between values and emotional expressions should be addressed on the individual "meaning-level" (p. 1, 5). In addition, Görnemann and Spiekermann consider emotional experiences and their related meanings to be situated and relational by nature, thus rejecting the possibility of producing universal descriptions for connecting certain emotional reactions with corresponding fixed meanings. Notwithstanding similarities of our approaches to moral sentimentalism, the framing of Görnemann and Spiekermann (2024) is set for user experience design purposes, and so uses a static view on emotions, and narrows aspects of moral thinking into consideration of values. As we argue in Chapters 2 and 3, neither one of these approximations is suitable for evaluating and framing appropriate design when considering automated emotion recognition in AI ethics. A difference in research objectives causes us to have similar presumptions but to employ distinct theoretical framings of emotions and moral thinking. Where Görnemann and Spiekermann enlarge the scope and methods of user experience design, we critically evaluate and reframe the methodological landscape of using automated emotion recognition in AI ethics. Thus, we consider the work of Görnemann and Spiekermann highly noteworthy but being part of a distinct research setting.

2. Present-day automated emotion recognition

Our first concern—the relevance and scientific validity of pursuits to develop automated emotion recognition in general—has been widely acknowledged. The literature can be roughly divided into studies depicting harmful social impacts (Boyd & Andalibi, 2023; Roemmich et al., 2023; McStay, 2020; Monteith et al., 2022; Nakamura, 2019; Stark & Hoey, 2021) and research on the scientific validity of the designs (Barrett et al., 2019; Branford, 2024; Heaven, 2020; Mohammad, 2022; Neal et al., 2019[15]; Stark, 2023). Both sides are important for addressing the question of relevance.[16] From an ethical standpoint, the development of harmful technology is not desirable; from the perspective of technological progress, technology based on deceptive assumptions is not desirable. These two aspects of relevance are closely connected. Monteith et al. (2022) have emphasized the social harm caused by automated emotion recognition applications that ignore individual differences in how people express emotions as well as the effects of this deficiency of the technology, especially for people who are already

marginalized, such as those with mental health conditions that affect their emotional information processing. Autism spectrum disorder (ASD), for instance, impacts how people express emotions and attribute affective dimensions to others. Therefore, using automated emotion recognition to evaluate a person's *emotional and social capabilities* (Ajunwa, 2022; Roemmich et al., 2023) in a hiring situation, for example, could introduce an excessive and biased negative effect for people on the autism spectrum.

While we notice the duality of relevance as represented above, this chapter concentrates on the aspect of scientific validity. Ethical considerations are addressed in more detail in Chapter 4 (Elaboration of limitations). At the heart of the validity discourse are underlying presumptions of the natural discreteness of emotions and their elicitation as temporally preceding the processing of other components of an emotional episode (Barrett et al., 2019; Branford, 2024; Keltner et al., 2023; Stark, 2023). These presumptions have led to an understanding that physiological reactions and physical expressions happen temporally after an emotion and are therefore *true* or *honest signals* (Gloor, 2022; Khare et al., 2024; Seetha et al., 2024; Zhang et al., 2024) of the person's mental state.[17] They have been implicitly presumed to justify the construction of models for the inference mechanisms of automated emotion recognition along rigid categorical paths that connect physiological reactions, physical expressions, or both to discrete emotions and even to the event that elicited the emotion (Branford, 2024). From an emotion theory perspective, the presumptions of temporality and discreteness fall under a classical view of emotions (Oatley et al., 2006; Russell, 2003; Scherer & Moors, 2019) or basic emotion theories (Cowen & Keltner, 2017; Cowen et al., 2019; Ekman 1992; Ekman & Friesen, 1971; Keltner et al., 2023). While they were already contested in the mid-20th century (Arnold, 1969; Arnold & Gasson, 1954; Frijda, 1986; Roseman, 1984), they received increased attention at the turn of the millennium (Ellsworth & Scherer, 2003; Oatley et al., 2006; Russell, 2003; Russell & Barrett, 1999). The most pressing controversy resulting from the discreteness and temporal presumptions within contemporary automated emotion recognition revolves around the idea of not having to consider emotional information processing from an intentionality point of view, as the need for such consideration is bypassed by implicitly accepting that the presumptions provide a path through such latent phenomenon.

In the design and use of sentiment analysis applications,[18] the temporality and discreteness presumptions have led to accompanying presumptions that emotion words, or emotion labels, are the central integration point of emotional information processing. Therefore, the analysis of textual communication is believed to reveal the *true mental states* of people. This claim has been made by automated hiring service providers, for example, in using sentiment analysis to assess the emotions and social capabilities of job applicants (Boyd & Andalibi, 2023; Roemmich et al., 2023; Ruckenstein & Velkova, 2019[19]).

In addition to the aforementioned presumptions relating to the operationalization of the *elicitation, emergence*, and *nature* of emotions as they appear in individuals, presumptions about the nature of natural emotion recognition and how it can be automized are a central aspect of automated emotion recognition, as many contemporary applications are presumed to follow the natural way in which people recognize the emotions of other people (Gloor, 2022; Hoegen et al., 2023; Khare et al., 2024; Kosti et al., 2020; Picard, 1997). On the one hand, psychological and cognitive science literature has considered natural emotion recognition as part of a person's capability to attribute internal states to other beings in order to predict their behavior (Brüne & Brüne-Cohrs, 2006; De Prisco et al., 2023; Folz et al., 2022; Leslie et al., 2004; Meinhardt et al., 2011; Scherer et al., 2018). On the other hand, it has been interpreted as part of the human need to attribute meaning to one's surroundings to enable meaningful interaction with one's environment. This need may lead to confident but invalid attributions (Barrett et al., 2019; Bonnefon et al., 2015; Malle, 2011; Todorov et al., 2015).

In much of contemporary automated emotion recognition, third-person evaluation occurring in natural emotion recognition is nonetheless considered a highly accurate process, which we call the precision presumption. The use of third-person evaluation is often referred to as crowdsourcing emotion annotation (Hoegen et al., 2023; Kosti et al., 2020). As crowdsourcing is considered a great way of producing labelled data for training machine learning models in general, the ability to estimate what can be crowdsourced and what cannot, can be considered an important operationalization blind spot, where AI approaches to emotions often go astray (Hoegen et al., 2023). The precision presumption has lead developers of the technologies to hold natural emotion recognition as a justifiable means of producing

training data for the machine learning models that the applications rely on. For example, when using image and speech recognition or text-based sentiment analysis techniques, machine learning models are trained to recognize emotion categories in most cases with data sets formed by forced-choice third-person annotation (Araño et al., 2021; Barrett et al., 2019; Khare et al., 2024; Ullstein et al., 2022; Yang et al., 2023). In some cases, the third-person evaluations are reinforced by forced-choice self-reported corrections[20] (Gloor, 2022). In the case of image (and sometimes speech (Araño et al., 2021)) recognition techniques, the third-person annotation is done by evaluating pictures of actors either in neutral settings or in staged *natural* settings that reveal the (physical) background context, often referred to as in-the-wild setting (Kosti et al., 2020). The emotion categories underlying the training data are mostly in accordance with basic emotion theories, which often recognize the existence of 4 to 12 basic emotions (Barrett et al., 2019; Cowen & Keltner, 2017; Ekman 1992; Ekman & Friesen, 1971).

In addition to justifying the use of third-person annotation of training data, the precision presumption combined with the presumption that people *detect* the emotions of others by observing their overt behavior has led to the simplification of the physiological and physical components of emotional information processing as *signals* of emotions through which the detection of emotions can be operationalized (Gloor, 2022; Khare et al., 2024; Seetha et al., 2024; Wei et al., 2024). The rise of these two presumptions (see, e.g., Wei et al., 2024) has been largely affected by an oversimplification of Ambady and Rosenthal's (1992) seminal work on the effectiveness of everyday layperson predictions of the consequences of interpersonal behavior. The most significant finding of this work is that people were reasonably accurate in predicting the consequences of interpersonal interactions when they had access to only a brief moment (maximum of 5 min) of an interaction taking place (e.g., in a video or sound recording). Similarly, Ekman's (1992, 1993) seminal work on basic emotion theory and facial expressions (Ekman 1992; Ekman & Friesen, 1971, 1978; see also Barrett et al., 2019) has been simplified as evidence that people are accurate in predicting mental states of other people. Despite how radical it sounds, the precision presumption has often led developers and deployers of automated emotion recognition to state that the evaluated person is lying if they do not identify the prediction[21] (Roemmich et al., 2023; McStay, 2020; Stark & Hoey, 2021; see also Zhang et al., 2023). This kind of argumentation implies a further presumption that emotional information processing is a static state,[22] and the *recognized* category resembles *the emotion* of a person.

All of the above-mentioned presumptions crucially omit mental representations and their contents and underpinnings from the central framework for understanding human information processing, thinking, and action. When attributing internal states to others, people process an immense amount of information beyond the physically visible context (including expressions) and physiological reactions of the people involved in the situation. Most importantly, people involuntarily evaluate latent information relating to the social context (i.e., the social roles of people and the expectations of behavior that are connected to those roles (Chen & Whitney, 2019; Goffman, 1956, 1979; Mobbs et al. 2006; Mumenthaler & Sander, 2015; Ross & Nisbett, 1991), social norms, and their violation (Malle et al. 2021; Monroe & Malle, 2019)). These aspects can only be very superficially interpreted from a third-person view since such latent information derives not solely from physically visible aspects but also from *historical* and *relational* factors for the people involved in the situation (Hoegen et al., 2023). People also evaluate possible involuntary appraisal patterns underlying emerging feelings, which involves interpretation of what has been appraised (Folz et al., 2022; Scherer et al., 2018). Even design suggestions of automated emotion recognition that are aware of the importance of contextual information in natural emotion recognition are highly limited. While they take some physical elements of the context into consideration (Branford, 2024; Hoegen et al., 2023; Kosti et al., 2020), they do not and cannot account for crucial latent social information. In addition to being based on latent information processing, natural emotion recognition often produces evaluations that are not fully correct but are *correct enough* for acting in everyday life (Barrett et al., 2019; Malle, 2011; Todorov, 2017; Todorov et al., 2015). Since people have a need to try to understand their surroundings, they form ideas of another person's internal state based on their own goals, capabilities, personality, and preceding experience. These considerations undermine the accuracy claims of any third-person evaluation of people's emotional information processing.

While the aforementioned presumptions inherent to much of contemporary automated emotion recognition are collectively explicated here and throughout previous critical literature, they are not well understood by developers and even less so by users of the resulting artefacts (Ajunwa, 2022; Barrett et al., 2019; Boyd & Andalibi, 2023; Branford, 2024; Hoegen et al., 2023, Roemmich et al., 2023; Stark, 2023). Currently, automated emotion recognition is pursued through image and video recognition (e.g., facial, posture, and gesture recognition) and speech recognition techniques (Lee et al., 2023; Zhang et al., 2023), the measurement of physiological responses (e.g., heart rate, skin conductivity, pupil size, cortical and subcortical electric potential) (Guo & Wang, 2024; Khare et al., 2024; Picard, 1997; Prabowo et al., 2023), and sentiment analysis of texts (i.e., language analysis) (Mohammad, 2022; Zhang et al., 2018). Notwithstanding the choice of technological solutions, the explicated invalid presumptions have led developers and users of these technologies to grossly overestimate their capabilities, stating that emotion recognition "is the ability to precisely infer human emotions from numerous sources and modalities" (Khare et al., 2024, p. 1), that people "could disguise their true emotions by disguising facial expressions and voice intonation," so emotion recognition based on physiological signals "can provide a more objective and authentic reflection of a person's emotional state" (Guo & Wang, 2024, p. 1), and that automated emotion recognition as "microexpression [detection] can express the true emotions that people are trying to suppress and hide" (Zhang et al., 2023, pp. 1, 7, brackets added). The following quotations further illustrate such overstatements:

> EAI hiring services position use of their technology as a moral imperative by characterizing EAI as the one true entity capable of solving organizational problems in hiring. (Roemmich et al., 2023, pp. 2, 9)

> Through hidden subconscious 'honest signals', body language, word usage, and facial cues, individuals signal to each other their emotional state … Emotions can be measured in many different ways, looking at facial expressions, body signals of how somebody moves, voice patterns such as the tone of the voice, and at the choice of words somebody uses when communicating with others. (Gloor, 2022, pp. 20, 140)

These examples show how the presented scientifically false presumptions connect the possibility of detecting and recognizing a person's *true* mental states to the design and use of automated emotion recognition. Truthfulness, in this case, is contrasted with the idea that people are intentionally deceptive or do not know their feelings. It is true that people can voluntarily produce expressions and, to some extent, even affect their involuntary reactions and expressions.[23] Likewise, it is true that people are not always able to fully capture some part of their affective experience and verbally express it (Dennett, 1988; Scherer, 2009). However, these considerations do not imply that automated emotion recognition relying on a third-person evaluation can access the mental content of a person any better than the person can themself. Such a belief is out of touch with the careful conclusions of Picard (1997), a pioneer in automated emotion recognition:

> Although I write of "recognizing emotions," I am not proposing that computers could recognize or measure affective states directly. Because affective states are internal and involve cognitive thoughts as well as physical changes, they cannot be fully recognized by anyone but the person having the affective state. Outsiders only have access to observable functions of the affective state—expressions, behaviors, and so forth. (Picard, 1997, pp. 166–167)

The overstatements relate to perceived explanatory power, which is a critical aspect in evaluating the ecological and external validity of the research, design, and use of automated emotion recognition. The question here is, "To what extent can automated emotion recognition be said to provide an explanation of the mental information processing of a person?" This question covers multiple, scientifically distinct conceptual realms (e.g., psychology, cognitive science, emotion studies, computation, engineering, artefact, and interaction design) and cannot be answered by simplifying other areas to fit a chosen perspective. We return to this *areas of reasoning* point of view in the next subsection. Here, we wish to show that the gross overstatements rely on the simplification of conceptual knowledge from other fields to fit the technical requirements of automated emotion recognition. Such simplification is highly visible in propositions such as "[e]motions are the external reflection of the inner mind, which plays an important role in communication" (Zhang et al., 2023, p. 1). This definition, constructed for engineering purposes, seems to involve kernels of truth but has no theoretical or empirical validity. Emotions have an important role in communication but are not an external reflection of the inner mind. The definition

misinterprets some components of emotional information processing—physiological responses and physical expressions and representations—as *being* emotions. This example reflects how the idea in automated emotion recognition research and design often reverses the interpretation processes of emotion theories. Emotion researchers have attempted to understand emotions and affective phenomena as holistic processes that are best explained by examining a person's mental information processing, which affects external visible behavior (Cowen et al., 2019; Power & Dalgleish, 2016; Scherer & Moors, 2019). Automated emotion recognition designers and deployers have failed to realize that reversing the flow by explaining the *inner mind* according to mere observations of external responses falsely simplifies a huge amount of theoretical and empirical knowledge about emotions.

The discreteness presumption, a central aspect of dispute within contemporary automated emotion recognition, can be traced back to theoretical disputes within emotion studies. However, emotion researchers from a variety of standpoints have likened such applications, which rely on detecting and evaluating physical and or physiological attributes of people, to pseudoscientific endeavors (Heaven, 2020). Even emotion researchers who lean on the discreteness assumption have not denied the differentiation of findings within discrete emotion categories, instead explaining it as epiphenomenal (Cowen & Keltner, 2017; see also Siegel et al., 2018). Whether such differentiation is epiphenomenal or not, researchers have considered it problematic for technologies that rely on the discreteness assumption since they are intended for use in real-life settings, where differentiation is inevitable (Barrett et al., 2019; Heaven, 2020).

Thus far, we have explicated the intertwined presumptions related to the temporality, discreteness, and static nature of emotions, emotion labeling, and the nature and preciseness of the third-person annotation inherent to contemporary automated emotion recognition. These presumptions have led to gross (but unnoticed) ecological and external validity issues when used in operationalizing emotional information processing and its (third-person) recognition. In totality, these presumptions reflect the implicit adoption of *the behaviorist model of explanation* and the *classical view of basic emotions theories*—two major ideas that have been rejected or highly contested in cognitive science and emotion studies but are still mistakenly considered valid in contemporary automated emotion recognition. The behaviorist model of explanation formulates explanations in accordance with a radical positivist and reductivist ontology, as evident in the work of B.F. Skinner (Graham, 2023; Skinner, 1974, 1977). This approach neglects the significance of internal mechanisms of the mind in explaining human behavior. Furthermore, it omits concepts referring to mental qualities. When omitting them is not possible, such concepts may have a place in the explanandum, while behavior is the explanans.

Given that the behaviorist model of explanation used within contemporary automated emotion recognition appeals to concepts and theoretical entities referring to latent mental states of a person that are not accepted in strict behaviorism, it is more accurate to characterize the model as one of concealed behaviorism or methodological behaviorism (Graham, 2023; Saariluoma, 1997).[24] However, it employs explanatory frameworks (Saariluoma, 2005) that consider only the stimulus-response dimensions of a person and therefore neglects the explanations based on multidimensional latent mechanisms used in scientific literature related to emotion studies (Barrett et al., 2019; Cowen et al., 2019; Keltner et al., 2023; Power & Dalgleish, 2016; Scherer & Moors, 2019; Scherer et al., 2018; Siegel et al., 2018). The narrowness of the explanatory frameworks is apparent in how key theoretical entities and concepts within emotion studies, such as multi-level appraisal of an event, are not part of the explanans but instead objects of explanation through the interpretation of external features (i.e., part of the explanandum). Considering that behaviorism as a plausible scientific paradigm was abandoned by the scientific community over 60 years ago (Miller, 2003), it should not be accepted as the guiding theory of the scientific paradigm for designing and applying automated emotion recognition.

The behaviorist model of explanation narrows down the ideas presented in basic emotions theories in a way that renders them scientifically invalid. This narrowing is evident when comparing the presumption in contemporary automated emotion recognition that expressions or physiological states are *true signals*, or analogical to experienced discrete emotions, and the explicated presumption underlying the work of Ekman (1993), who helped found the basic emotions approach. Ekman (1993) stated, "I have reverted to facial expression because it is more felicitous, although it should be clear that in my view (Ekman, 1977) expression is a central feature of emotion, not simply an outer manifestation of an internal phenomena" (p. 384).

Considering basic emotion theories from a behaviorist viewpoint, current approaches to automated emotion recognition often transcribe epistemological commitments of basic emotion theories (we *can have meaningful information* about a limited set of emotions by observing expressions of people) into ontological ones (the observed expressions *are* a direct, causal link to mental states of people, and expressions can thus be considered emotions) (e.g., Bernstein, 1983). Furthermore, we do not dispute that basic emotion theories offer important ideas and results, such as a focus on emotion words, for the study of emotions. However, when adopted as the sole explanatory framework for emotional information processing, basic emotion theories are overly in contrast with a vast number of empirical findings (Barrett et al., 2019; Kreibig, 2010; Scherer & Moors, 2019; Siegel et al., 2018) to produce valid models for automated emotion recognition. Some seminal researchers, such as Cowen and Keltner (2017), still believe that basic emotion theories can inform explanatory frameworks, but their arguments largely depart from the principles of previous basic emotion theories by specifying a much broader set of basic emotions (27 rather than the usual 4 to 12), and they relax the notion of drawing a precise line between discrete emotions by more explicitly introducing the idea of the continuity of emotion spaces (Keltner et al., 2023).[25]

Because of the invalid presumptions underlying much of contemporary automated emotion recognition, some researchers have interpreted all automated emotion recognition as a modern implementation of physiognomic or phrenological ideas (Ajunwa, 2022; Stark & Hutson, 2022). While this stance is justified for the cases presented in this chapter so far, the following subsection discusses possibilities for scientifically valid automated emotion recognition to balance these extreme viewpoints.[26] As part of correction suggestions, we note that the term *recognition* implicitly bears the problematic idea of the *precision* of the technology, and thus, we henceforth employ the term *automated emotion inference* (AEI) to refer to pursuits of building technical artefacts for the purpose of sensing and interpreting people's emotional information processing, as suggested by Barrett et al. (2019).

2.1. From behavior to action

It is important to emphasize that the depicted behaviorist approach to emotions that is present in much of contemporary automated emotion inference *is not* a reflection of any existing psychological theory of emotion, which underlies its scientific invalidity. In moving forward with our investigation of the scientific relevance of automated emotion inference design and development, we note the careful unanimity among researchers who have acknowledged the current difficulties but remain in favor of developing the field. These researchers have posited that a valid design and use could enrich our understanding of the role of human emotions in human-technology interaction, thus supporting the creation of more meaningful technologies (Barrett et al., 2019; Gratch et al., 2024; Jeon, 2017; Mohammad, 2022; Picard, 1997; Scherer et al., 2021).

We start the search for valid premises for automated emotion inference by discussing cognitive approaches to emotions that consider cognitive and affective systems as interactive information processing systems within a single organism (Mischel & Shoda, 1995; Power & Dalgleish, 2016). Cognitive functions, such as perception, attention, memory, language use, decision making and problem solving, allow people to attain and process information from their environment in the first place. Affective processing determines the relevance of the attained information for the individual on multiple levels, which then influences which information is further processed and attended to (Damasio, 2006; Frijda, 1986; Oatley et al., 2006; Power & Dalgleish, 2016). Therefore, human thinking always integrates the two systems, and strictly separating them on a theoretical level easily leads to invalid hypotheses and argumentation (Damasio, 2006; Paavilainen, 2020), especially when interpreting and inferring human thinking not in a laboratory but in a real-life setting (Barrett et al., 2019; Saariluoma, 1997; Scherer & Moors, 2019), which is typically the case for automated emotion inference deployment (Ajunwa, 2022; Boyd & Andalibi, 2023; Mohammad, 2022; Picard, 1997).

The concepts of *mental representation*, *apperception*, and *appraisal* are key for further analyzing the integration of affective and cognitive systems within emotional information processing. Mental representations form the core of the cognitive theory of mind: they explain why people do what they do. The information content of representations (i.e., mental content) differentiates between courses of action.

However, many aspects of mental content are not directly linked with perception (Churchland et al., 1994), and non-perceivable kinds are often essential to the content of representations. People do not perceive entities such as eternity, infinity, possibility, events of tomorrow, or general ideas (e.g., car likeness). Perceivable and non-perceivable content elements must be integrated in the construction of mental representations. In this sense, it is appropriate to follow Kant in differentiating between *perceiving* the environment and *apperceiving* mental representations (Kant, 1781; Leibniz, 1704; Saariluoma, 1990, 2003).

People not only represent cognitive content in their mind but also integrate emotional dimensions of actions, events, and objects into the information content of mental representations. The process of integrating emotional information has been termed *appraisal* (Moors et al., 2013). As a whole, appraisal is, analogical to perception, a *subprocess* of apperceiving. Understanding apperception and mental content makes it possible for researchers to comprehend how the mind can control and guide human actions (Bandura, 1991; Saariluoma, 1997; Wallach et al., 2010; Zhu & Thagard, 2002). Cognitive approaches to emotions require changing from the central conceptual schema of behavior, which considers people as passive beings reacting to external stimuli, to one that views emotional information processing as part of people's active engagement with their environment (i.e., people as *action*-taking beings).[27]

In considering automated emotion inference as part of a possible moral competence mechanism, it is notable that morally relevant mental content, such as internalized ethical and legal principles, enables people to decide between various courses of action. Some deeds are interpreted as good, while others are interpreted as having bad consequences. Thus, apperception is a central mechanism in how people decide between good and bad, as such differences are not directly visible. Apperception also illustrates how emotions and cognition are integrated to allow for evaluating valuations (i.e., emotional values) (de Sousa, 2022) of the consequences of actions and forming ethical principles on the basis of such knowledge (Saariluoma, 2021).

In this vein, appraisal theories of emotions are pivotal when searching for ways to operationalize automated emotion inference. Currently, the event appraisal-initiated component processing model of emotions (Scherer, 2009; Scherer & Moors, 2019) provides the most detailed description of how involuntary appraisal mechanisms elicit higher-level emotional information processing in people and how the components interact to form a dynamically unfolding emergence of emotional information processing. It is a promising yet complex theoretical framework for automated emotion inference. After discussing the model in more detail, we consider its potential for automated emotion inference with respect to two important design concepts—*emotion task* (Mohammad, 2022) and *inherent areas of reasoning*—that must be present for the operationalization of valid automated emotion inference.

Appraisal theories are the most visible cognitive approaches within emotion studies (Jokinen & Silvennoinen, 2020; Moors et al., 2013; Power & Dalgleish, 2016). A common feature of appraisal theories is that the evaluation (appraisal) of an individually significant event is the main mechanism of emotion elicitation. The most distinctive proposition of the component processing model is a dynamic architecture for describing the effects of (1) *sequential multi-level appraisal checks* of an (2) *event* on the synchronization of the response systems of (3) *motivational changes* (i.e., action tendencies), (4) *physiological response patterns* (e.g., motor expressions), and the (5) *central representation of all components* (i.e., mental representation of feeling), which occasionally leads to the (6) *categorization* and possible (7) *verbal labeling* of the experience. Each of these numbered components has a different relation to emotional information processing. It is notable, however, that the mental representation of feeling is considered the central integration point of emotional information processing (Scherer, 2009).

The sequential checks are composed of four chronologically ordered levels with multiple criteria:

1. *Relevance*—Criteria: novelty, valence (as intrinsic pleasantness), and goal relevancy
2. *Possible implications*—Criteria: agency (initiator [object, subject] and intentionality), outcome probability, discrepancy from expectation, conduciveness to goals, and urgency
3. *Coping* (potential)—Criteria: control (if the situation can be controlled in general), power (if I can exert control), and adjustment (if I can adjust to the consequences if control is not possible)
4. *Normative significance*—Criteria: internal standards compatibility and external standards compatibility.

Together, these appraisal checks comprise a dynamically unfolding and subjective assessment of an event, its consequences and implications, and possible responses. Each sequential check affects the information processing that occurs in other components of emotional information processing, including motivational changes, physiological response patterns, the central representation of all components, and the (possible) categorization and verbal labeling of the experience, which in turn further feeds the evaluation process of the subsequent sequential checks. Thus, the components are highly interactive with one another. The strengths of the component processing model are its ability to explain individual differences in emotion elicitation and its understanding of emotional information processing as a dynamically unfolding process (see Scherer, 2009, p. 1308 for a further illustration of the model).

As emotion categories and the dimensional features of emotions are key concepts in contemporary automated emotion inference, we briefly discuss their significance to the component processing model. Generally, in cognitive approaches to emotions, emotion categories are considered individually apperceived schemes or scripts (Power & Dalgleish, 2016; Russell, 2003; Saariluoma, 2003; Scherer, 2009) that underlie the process of experiencing an emotion feeling as a discrete emotional state. Despite the importance of individual differences in emotional information processing, we share many commonalities in living conditions, interactions with the environment, and biological origins with other people in our cultural spheres—and, to a meaningful extent, also over the immediate spheres—so our emotional experiences have similar features to those of other people. Scherer (1994, 2009) has referred to the most typical similar emotional experiences as resemblances of *modal emotions*. We can produce abstracted thematic descriptions[28] for known and documented categorizations of emotional episodes (Power & Dalgleish, 2016; Saariluoma, 2003). From the perspective of the component processing model, the *theme*, or *mental content* of a feeling, is constituted by the combination of the underlying appraisal results. Since there is an almost infinite number of possible combinations of appraisals across the different levels of sequential checks, there is also a nearly limitless number of possible emotion themes (Scherer, 2009). The most highly abstracted theme descriptions do not describe emotions exactly as they are experienced; rather, they provide an opportunity to discuss family resemblances of experiences and to differentiate between certain experiences within the same family by further nuancing their appraisals.

Emotion themes further involve *what* is appraised, *how* and *why* it is appraised, and *what implications* the appraisal has for an individual's mental representation. These inquiries concern mental content and are of utmost importance to understand the emotional information processing of an individual (Myllylä & Saariluoma, 2022; Oatley et al., 2006; Power & Dalgleish, 2016; Saariluoma, 2003; Scherer, 2009). Thus, such questions can be answered by analyzing the mental content related to a feeling. We all intuitively understand that anger, fear, and sadness represent different emotions, but explicating the information content of the feelings corresponding to those emotional concepts can clarify *how*. However, the explication of single dimensions is not enough, as, for example, it is commonly held that all negative[29] emotions are appraised in some way as incompatible with existing goals, the concept of self, the model of others, or the model of the world (Power & Dalgleish, 2016).[30] Therefore, more information is required to determine if a person is angry, fearful, or sad. There is a highly abstracted way to distinguish between these feelings: anger involves a *judgement of blame* on someone for *intentionally* causing a negative event (i.e., the attribution of intent to a person who committed a wrongdoing); fear entails an appraisal of a threat (i.e., the chance of future non-completion of valued goals and small possibilities to avoid the realization of the threat); and sadness encompasses an appraisal of the loss of something valuable.

Multiple emotion themes may be present simultaneously. For instance, a person can be both sad and angry, which indicates that they are placing blame on another person or thing for the loss of something valuable. Furthermore, a single theme may include multiple emotions; for example, nostalgia is usually characterized by both sadness that some aspect from the past is no longer present and happiness about the good things it yielded. The explication of themes may also reveal important nuances within a single emotion family. In the case of fear, one may speak of fear of an encountered bear (i.e., a highly urgent threat to life), fear of the dark (i.e., a looming threat and incompetence to respond to it), fear caused by a noise in the dark (i.e., a novel sound in that context and the threat implied by its source and

implications), or fear of losing a game (e.g., the threat of *losing face*). Thus, the concept of emotion themes can help to explicate family resemblances or abstract analogical resemblances as well as describe important nuances within a given family of emotions (Cowen et al., 2019; Frijda, 1986; Keltner et al., 2023; Power & Dalgleish, 2016). As the examples show, in addition to differing interpretations of an event, different themes and subcategories can be distinguished by appealing to central, (often) dimensionally described phenomenal aspects present in all emotional experiences, such as novelty, valence, arousal, and dominance (Plutchik, 1980; Russell, 2003; Scherer, 2009). The component processing model considers that dimensional experiences are not only produced but also interactively and iteratively affected by the underlying appraisal patterns. Thus, appraisal patterns are the main means for understanding the encoding (i.e., elicitation and emergence) of emotional information processing on the level of emotion themes or dimensions (Power & Dalgleish, 2016; Scherer, 1994, 2009).

Notably, emotion themes provide a dense format for describing and communicating about highly complex levels of emotional information processing for people in everyday life (Russell, 2003) as well as for emotion researchers (Cowen & Keltner, 2017; Frijda, 1986; Power & Dalgleish, 2016). The operationalization of emotion themes has therefore gained wide interest in the field of automated emotion inference. However, emotion researchers apperceive the kinds of phenomena that underlie the *encoding* (i.e., emotion elicitation and emergence) or *decoding* (i.e., third-person inference of emotion elicitation and emergence) of emotion themes. If one lacks the expertise to understand such complex underlying mechanisms, then the operationalization of the central concepts will rest on a superficial understanding of them, which can lead to setting invalid limitations for operationalization possibilities or for the appropriate evaluation and use of constructed artefacts (Neal et al., 2019). Because of the density of emotion themes as an explication form, they can cause an overly simplistic understanding of emotional information processing. For example, contemporary automated emotion inference development does not recognize that the phenomenon of categorizing feelings in an individual's mind (i.e., the themes) and the process of using emotion labels to communicate the experience (i.e., searching for conceptual and verbal representations analogical to the experienced theme) are distinct components of emotional information processing that rely on different psychological mechanisms (Barrett et al., 2019).

Emotion words (or labels) are ways for individuals to convey their internal dynamic processes to others. Prior to labeling, individuals must categorize their experience in some way to be able to explain aspects of it. Categorization happens according to more or less complex internalized emotion schemas that arise associatively within the emotion elicitation (Frijda, 1986; Power & Dalgleish, 2016; Russell, 2003; Scherer, 2009). Since the categorization emerges as part of the (at least partly) ineffable feeling component of an emotional episode, it is not easy for a person to grasp the underlying involuntary appraisal processes; therefore, self-report of emotions has limitations in explicating the categories, dimensions, and underlying mechanisms of emotional information processing (Dennett, 1988; Scherer, 2009). In addition, since emotion words are part of languages, which are cultural constructs, they serve to describe certain attributes of an experience that an individual finds important to communicate, but they hardly ever convey the experience in totality (Mohammad, 2022; Scherer & Moors, 2019). In this way, emotion words are part of an individual's information processing and part of our cultural constructs. Internalized prototypical emotion categories (or emotion schemas) as apperceived by an individual are distinct from prototypes produced by analyzing aggregated language use of groups of people, though there is a mutual influence between them. This perspective is important for automated emotion inference because it clearly frames text-based approaches as an analysis of emotion words and their role in our languages, especially since such approaches currently rely on aggregated population-level data instead of individual data (Chen et al., 2023; Yang et al., 2023). Population-level data are valuable for gaining a holistic view of affective dimensions in discourses but not for the analysis of an individual's mental state, which is presumed to be the case when such technology is used for profiling individuals in, for example, hiring situations or education performance evaluations (Ajunwa, 2022; McStay, 2020; McStay & Rosner, 2021; Mohammad, 2022; Roemmich et al., 2023; Ruckenstein & Velkova, 2019).

From the perspective of attribution theories (Malle, 2011; Todorov et al., 2015) and theory of mind (Brüne & Brüne-Cohrs, 2006; Meinhardt et al., 2011), emotion words are significant in how people

communicate and understand emotions in everyday life. As such, the emotion words found and used in basic emotion theory approaches provide meaningful material for sentiment analyses of texts (Mohammad, 2022; Saariluoma & Jokinen, 2014). Our position on text-based automated emotion inference overall is in accordance with Scherer and Moors (2019), who have acknowledged the use of emotion labels (or emotion words) as an important part of people's emotional information processing—but without considering it an inevitable part of the process, let alone the central integration point. The same limitations apply to text-based automated emotion inference as to the use of verbal self-reports in emotion research: there are limits to what can be analyzed from verbal expressions alone. In the development and use of text-based approaches, emotion words should not be viewed as emotions themselves but rather as verbal representations and cultural, learned constructs that individuals use to express dynamic multidimensional experiences (Mohammad, 2022; Russell, 2003; Scherer, 2009; Scherer & Moors, 2019; Siegel et al., 2018). Such analysis should not be seen as analyzing *the true mental states* of people; instead, it should focus on how a certain text compares to our understanding of how emotions are expressed and communicated in our language in general.

In the following section, we further discuss how the key design concepts of the *emotion task* and *inherent areas of reasoning* can be used to incorporate views from cognitive approaches to emotions into automated emotion inference activities.

2.1.1. The emotion task and inherent areas of reasoning

The construct of an emotion task (Mohammad, 2022) in automated emotion inference design clarifies from the outset of the design process that a third-person evaluation cannot fully recognize a person's mental state but *might* be able to recognize some aspect of a person's emotional experience at a single point in time in a clearly articulated context (i.e., a dimensionally limited state). Here, the use of the term "emotional state" does not suggest that emotions are a static phenomenon; instead, it indicates that the foundations of automated emotion inference technologies cause them to be capable of performing analysis only by observing certain aspects of a single point (i.e., state) in a person's emotional information processing. As such, automated emotion inference should always have a clearly articulated role in a well-framed emotion task that is characterized by one or more explicit goals in a well-known context. In this sense, emotion task is a concept that points to the inherent limits of artefact design, just as the concept of moral competence does in AI ethics (Malle & Scheutz, 2019). Both concepts emphasize that there is no *one-size-fits-all* solution; on the contrary, all valid artefact instantiation processes must start by framing and analyzing the situation that is meant to benefit from the development and use of the technical artefact (Gillespie, 2019; Salo-Pöntinen, 2021). The design questions that should be answered in the design process of the technical artefact can be determined by explicating the characteristics of the overall situation and the perceived task (Saariluoma et al., 2016).[31]

Thus, the design of a validly automated emotion task is always the end result of a careful and contextually laden analysis "avoiding careless endorsement of theories that suggest a mapping of external appearances to inner mental states" (Mohammad, 2022, p. 245). A crucial part of framing an emotion task is to analyze the task context in physical, psychological, social, and cultural terms and carefully specify the goals of possible tasks[32] within that context. Afterward, one may select automated emotion inference methods that fit the chosen goal(s), or vice versa, to frame the task goal(s) in congruence with the methods.[33] This step sounds intuitive but is often the point at which automated emotion inference development goes astray. For example, language-oriented automated emotion inference is largely based on "[i]nferring emotionality of language used in text (regardless of whose emotions)" (Mohammad, 2022, p. 244). Therefore, while sentiment analysis is suitable for evaluating aggregated attitudinal data (positive, negative, or neutral) from large groups of people toward some evaluated entity, a significant number of sentiment analyses have set the identification of the emotional information processing of individuals as the main goal (Gloor, 2022; Mohammad, 2022; Roemmich et al., 2023; Ruckenstein & Velkova, 2019).

Emotion tasks are formed to fit the context. The context is the physical, psychological,[34] social, and cultural environment in which the task takes place, whereas the emotion task is defined by the goals of the automated emotion inference activity taking place in the context. Hence, the context and the

emotion task are not interchangeable, and multiple emotion tasks can be conducted in the same context. For example, a text-based social media platform can be the context for inferring the attitudes of a group of people toward a single entity or for inferring which kinds of emotions people seek to convey in discussions in general. It is important to note that all possible emotion inference methods have particular benefits and limitations, the effects of which have to be evaluated against the framed emotion task happening in the given context. In this way, *the nuances* of the emotion task that are *relevant for evaluating* the end results of using automated emotion inference *are revealed*. These nuances must be understood and communicated to all stakeholders, as "[n]ot doing so will lead to the misuse and misinterpretation of one's work" (Mohammad, 2022, p. 248).

We provide two guiding questions to help explicate the underlying presumptions and task-relevant sub-questions when designing emotion tasks. First, as stated, automated emotion inference activity is always a process of third-person evaluation. Therefore, when forming goals for emotion tasks, the first question is, "in the task, what is evaluated from a third-person perspective: textual expression, overt expressions and responses, or both?" Second, automated emotion inference has the potential to base third-person evaluations on our knowledge of emotion encoding (i.e., how emotions emerge in the information processing of people) or decoding (i.e., how people evaluate the emotional information processing of others). Therefore, the second question is, "is the task based on knowledge of encoding or decoding?"

The answers have far-reaching consequences for the analysis of limitations of using automated emotion inference (e.g., Barrett et al., 2019; Mohammad, 2022). A cognitive approach to emotional information processing must consider that the appraisal of the eliciting event is developed through apperceptive processes (Saariluoma, 1990, 1997) that underlie the formation of all mental representations. Thus, even if the event is comprised of a multitude of properties that are not necessarily subject to the individual perceiving them (i.e., objective properties of the context), the subject's object of appraisal is formed by attaching individual relevance (i.e., narrowing down *what is* perceived) and meaning (i.e., *how* it is apperceived) to the event under inspection, and, in some cases, the event is formed completely in the person's mind without any connection to the external environment (i.e., through thoughts and memories). In this sense, if there is no access to mental content-related information (i.e., textually expressed information), the third-person evaluation cannot clearly establish *what* elicited the emotional information processing or definitely identify the content of the processed mental representations (Myllylä & Saariluoma, 2022; Saariluoma, 1990).[35]

Is there any possibility then, even in theoretical terms, to develop valid automated emotion inference artefacts that base their inference on observing overt expressions or responses (hereafter "overt automated emotion inference")? This question brings us back to evaluating the explanandum and explanans in automated emotion inference. If we can have knowledge of cognitive mechanisms that have been empirically proven to reliably lead to similar overt responses on a relevant population level, and if the knowledge can be included in the automated emotion inference artefact to provide an explanation of the observed overt responses,[36] then we can cautiously claim to have premises for such automated emotion inference. The next example shows that, when speaking of arousal (i.e., the intensity of emotions), this explanation integration is known to be achievable by, for example, examining indicators of individual changes in heart rate and respiratory activity. Therefore, it can be said that it is possible to detect when something has prompted the emotional information processing of people (Scherer & Moors, 2019). However, this does not imply that the same means would provide any access to the *information content* of said information processing.

In the last decade, two meta-analysis articles covering findings from a combined total of 336[37] studies on autonomic nervous system and somatic nervous system reactions to specific discrete emotions (Kreibig, 2010; Siegel et al., 2018) have indicated that the discreteness presumptions of basic emotion theories, or the classical view, are not supported by empirical evidence (see also Barrett et al., 2019; Scherer & Moors, 2019). These findings have led researchers such as Barrett et al. (2019) and Siegel et al. (2018) to suggest abandoning basic emotion theories altogether. In a less extreme example, Scherer and Moors (2019) have appealed not to discrete high-level emotions but to the effects of lower-level component processing explanations for physiological and physical responses that take place in

affective phenomena. The appraisal of an event on five interlinked levels[38] and the combined effect of the evaluations produces action tendencies that then steadily affect the increase in (or calming of) the heart rate and respiratory activity and the rise in the feeling of valence (Shuman et al., 2013). The physical reactions of a rising heart rate and increased respiratory activity indicate a state of arousal; however, since emotions with different valence content may nonetheless involve similar arousal states, heart rate and respiratory activity cannot be held as indicators of valence or other relevant emotional content. This example shows that when higher levels of emotional information processing are analyzed, simplistic solutions are bound to fail. According to Scherer and Moors (2019), since appealing to discrete emotions diminishes the credibility of empirical evidence, empirically valid emotion research ought to consider the appraisal-initiated component processing level as the main premise for explanatory frameworks of emotional information processing.

The discussions we have engaged with so far evoke doubt about the possibility of overt automated emotion inference from the perspective of the *mental content* of emotional information processing. As noted, the required reliable empirical evidence has not been provided on the level of discrete emotions (Kreibig, 2010; Siegel et al., 2018). Yet, the approach of the component processing model can enable a discussion of the issue from the viewpoint of lower-level appraisal mechanisms, as Scherer and Moors (2019) have suggested. As an interesting starting point for such investigation, Barrett et al. (2019) have evaluated a massive amount of facial expression experiments and found no evidence of the existence of discrete emotions as they are understood in the basic emotion theory tradition. However, their findings suggest that when changing the frame of analysis to evaluate the results as indicating an evaluation of *valence* (i.e., experiencing a situation as positive, negative, or neutral), the results were seemingly consistent across the reviewed experiments.

This finding of Barrett et al. (2019) is intriguing. However, since the experimental settings they examined were not designed for examining valence, the researchers imposed their own view of valence on all of the experiments. They perceived valence as an irreducible experiental building block[39] present in all human experience that refers to the quality of an individual's experience on a positive-neutral-negative spectrum. This view of valence is founded on experimental presumptions that take the experiential aspect (i.e., conscious experience) of emotional information processing as the main point of analysis.

It is, however, possible to examine appraisal mechanisms that lead to the experiential entity of valence, which Barrett et al. have considered irreducible (Scherer, 2009; Scherer & Moors, 2019; Shuman et al., 2013). This perspective approaches valence on two levels: micro and macro. The micro-level refers to qualitatively distinct event appraisal criteria that are valenced (i.e., their outcome is considered on the spectrum of positive-neutral-negative). Such appraisal criteria are *intrinsic pleasantness*, *goal conduciveness, power* (i.e., feeling of control), *self-congruence*,[40] and *moral goodness* (Frijda et al., 1989; Shuman et al., 2013). At the macro-level is the valenced *feeling* that results from the combination of the lower-level valence appraisals (Shuman et al., 2013). The division between the micro- and macro-levels helps to explicate why even the experience of valence may involve contradictory elements[41] (Gentsch et al., 2020). For example, aspects of a situation that may be appraised as intrinsically pleasant may also be appraised as incongruent with one's self-concept. Shuman et al. (2013) have stated that, by providing important nuance for how valence is considered, the micro-level of valence gives a better understanding of people's emotional information processing in a given situation, but the macro-level approach may be fruitful for examining how the integrative feeling of valence helps people compare alternatives in decision-making settings.

Recent experiments relying on the component processing model have shown that some of the (temporally speaking) first micro-valence appraisals (i.e., perceived inherent pleasure and goal congruence/conduciveness of the appraised event) have steady distinctive appearances in the muscular activity of the zygomaticus (cheek) and corrugator (brow) areas and in the heat temperature of the forehead (Aue et al., 2007; Aue & Scherer, 2008, 2011; Gentsch et al., 2015, 2020; Scherer et al., 2017; Scherer & Moors, 2019; Wu et al., 2012).[42] In these experiments, the appraisals of single criteria under examination were only one small part of a person's emotional information processing, and the same appraisal result for a single criterion (e.g., goal congruence) could be part of almost an infinite number of different appraisal constructions (i.e., the consciously experienced representations in which the dynamic and unfolding appraisal

results are iteratively integrated). Therefore, the experiments support the presumptions of the component processing model but not the discreteness presumptions of basic emotion theories (Scherer et al., 2021). According to the component processing model, the results of only a single appraisal criterion offer very little knowledge about the whole emotional information process (Scherer, 2009).

These findings seem theoretically promising for overt automated emotion inference. However, it is important to note that the experiments were conducted in laboratory settings, and the findings rely on the use of electromyography to detect visually imperceptible changes in muscular activity (Gentsch et al., 2020; Scherer & Moors, 2019). Therefore, while the results suggest theoretical possibilities for constructing valid overt automated emotion inference, *the results can in no manner be straightforwardly converted into part of automated emotion inference artefacts*, which has been the mode of action in contemporary approaches. More experiments are required to confirm the findings, and a vast amount of *field experiments* (i.e., experiments in non-laboratory settings) should also be conducted to test the ecological validity of such automated emotion inference pursuits. However, even in field experiments, it is likely for eliciting events to be controlled; thus, even field experiments cannot definitively determine what the eliciting event in a given moment is—a question that is always present when using the third-person view to make inferences about emotional information processing in real-life settings. In addition, since electromyography is an invasive means of measurement that is obstructive in normal life settings, there are technical barriers to consider, such as whether it is possible to construct machine learning programs that can detect similar muscular activity in image materials that is detected with electromyography in a laboratory setting (Zhang & Arandjelović, 2021).[43]

These validity questions bring us to the theme of *areas of reasoning*. The idea of framing areas of reasoning inherent to the operationalization of valid automated emotion inference is established here to emphasize the need to actively avoid theoretical naivety in such activities. Layperson explanations are not considered valid in scientific activities (Kaplan, 1964; Saariluoma, 1997). However, the *lending* of theories, concepts, and explanatory frameworks from other fields that often occurs in the design of automated emotion inference is guided by a layperson understanding that simplifies them in ways that are scientifically invalid (Barrett et al., 2019; Heaven, 2020). Behind this simplification is not the careful operationalization of the central empirical concepts or aspects of the phenomenon under inspection but rather the formalization requirements to convert the phenomenon into a computable form (Saariluoma & Karvonen, 2024; Salo-Pöntinen, 2021; Stark, 2023) and requirements set by the data gathering methods that are possible with the help of selected technical artefacts (Barrett et al., 2019). Thus, it is not sufficient to just acknowledge the existence of the theoretical entities; it is necessary to have deep expertise to *use* them properly. The centrality of field relevant expertise shows how the operationalization of the concepts and phenomenon is often the point at which the good intentions of designers go wrong, as they do not acknowledge limits of their own expertise and capacity to achieve a not only technically but also scientifically valid operationalization. For valid technology design and development, both technical and scientific validity are required (Barrett et al., 2019; Branford, 2024; Mohammad, 2022; Roemmich et al., 2023; Stark, 2023).

This insight about the difficulties of operationalization is vital because it reveals a complex point of junction that is generally insurmountable in the design of intelligent technologies. The design of intelligent technologies is always a case of multidisciplinary effort, and it may be difficult to recognize the boundaries of one's own knowledge (Salo-Pöntinen & Saariluoma, 2022). However, as our technologies are becoming more and more sophisticated and being used in widely varying contexts, the valid design of AI solutions is increasingly requiring the collaboration of experts from the humanities, social, behavioral, and natural sciences, and engineering. The context of design sets the demands for expertise from these fields. Such multidisciplinarity is not just a virtue—it is a necessity (Salo-Pöntinen & Saariluoma, 2022; Stark, 2023).

Operationalization is the point at which a critical understanding of justified use contexts and emotion tasks and the limits of the produced artefacts can be formed. Such understanding should be the basis for recognizing the artefact as valid for its purpose (e.g., Mohammad, 2022; Stark, 2023). As observed, in most cases of automated emotion inference, the operationalization of emotional information processing is constructed by technical boundaries to which other relevant areas of reasoning must submit.[44] To counterbalance this narrow understanding, we propose the design concept of *inherent areas of reasoning* to capture the different scientific and theoretical backgrounds that are involved in

explaining—and therefore operationalizing—the theoretical and empirical entities when developing any form of AI artefact. We further identify and describe the inherent areas in the case of scientifically valid automated emotion inference. The first is the area of reasoning that results in understanding the operationalization needs of the contextually laden affective phenomenon. Second, there is the area of reasoning that explains the technological capabilities possible for the operationalization. Third, there is the area of reasoning that explains what it means to form an integrated instantiation of the two previous areas of reasoning. This third area of reasoning manifests in the idea of an emotion task (Mohammad, 2022). It is a type of expertise that must increase in order for purposeful technological progress to be possible in an era where many socially and societally important functions are being transferred to AI artefacts (Deguchi et al., 2020; Eitel-Porter, 2021).

In this section, we have discussed theoretical and technical barriers to making inferences about emotions that must be acknowledged in valid text-based and overt automated emotion inference artefacts. The main point made here is that, from a scientific validity perspective, even in the best-case scenario, automated emotion inference may only provide hints about the emotional information processing of a person. Notable difficulties with detecting the eliciting event and the latent social context will remain even if overt automated emotion inference is developed according to the promising findings on micro-valence. In the next chapter, we discuss the potential of using automated emotion inference as a means of evaluating the moral dimensions of people's thinking.

3. Automated emotion inference and moral sentimentalism

In the past decades, moral psychology has been divided by the question of the *mark of the moral* (Hindriks & Sauer, 2020; McAuliffe, 2019), which considers the qualities that distinguish moral thinking and thoughts from non-moral ones. The two main lines in answering this question are the *rationalist* and *sentimentalist* perspectives. Proponents of a rationalist approach define the mark as being propositional in nature (i.e., that the contents of thinking that can be formed into propositions are the main way of defining if a thought is moral or not). In contrast, advocates of sentimentalism hold that it is not the content of propositions but the sentiments attached to them[45] (which either cause the formation of propositions or are caused by evaluation of propositions) that mark a thought as moral (Haidt, 2001, 2008; Kauppinen, 2022; Malle, 2021). Although this discussion has been dominated by disputes between sentimentalists and rationalists, neither side has denied that reasoning and emotional processes play a role in moral thinking. Rather, the debate is about which aspect has primacy.

For our purposes, it is important to explore the conceptual understandings linking emotional information processing and moral thinking. Such conceptual understandings have mainly been developed in literature with a sentimentalist tendency (Kauppinen, 2022). However, we do not try to provide an exact answer to the location of the mark of the moral or state primacy of one of the approaches. We instead note that, when considering human thinking from a holistic perspective, the most plausible approach is a hybrid form of rationalism and sentimentalism that emphasizes the integration of cognitive and affective effects as the main means of constructing moral thinking, as one cannot exist without the other. Such a dynamic view has ecological validity over considering single events within the formation of moral dimensions of thinking (Holtzman, 2018; Kauppinen, 2022; Liu et al., 2022; Scherer, 2009; Scherer & Moors, 2019; Schwartz, 2012).

We distinguish four conceptual levels for linking emotional information processing and moral thinking:

1. *Valuing* as instantiated in event appraisal results and component processing (de Sousa, 2022; Wallach et al., 2010)
2. The presence of *categorical moral emotions* as indicators of the moral nuances involved in an event (Haidt, 2003; Russell & Giner-Sorolla, 2011; Tangney et al., 2007)
3. The emotional response (attitude) as an indicator of *congruence or incongruence with internalized value systems* (Haidt, 2001; Rydenfelt, 2015; Schwartz, 2012), and
4. Social discourse as *value refinement* based on valuing(s) (Habermas, 1995; Nussbaum, 2004; Saariluoma & Rousi, 2020).

The different levels of appraisal proposed in the component processing model (Scherer, 2009) reveal that all of these conceptual levels have a multifaceted and dynamic effect on the emotional information processing of a person within single events.

However, the four levels have different development mechanisms from an ontogenetic point of view as well as different relations to the social context of emotion elicitation. While the valuation of some entity instantiated in a single event may change when the interpretation of the situation changes (Liu et al., 2022; Monroe & Malle, 2019; Power & Dalgleish, 2016), the internalized schemas launching the categorization of *moral* emotions of an individual require a longer time period and accumulation of experiences to be formed or changed (Kuppens et al., 2007, 2008; Oatley et al., 2006; Power & Dalgleish, 2016; Scherer, 2009). In addition, in the same way that emotion themes (or categories) are a high-level phenomenon within emotional information processing, the values of an individual are a high-level concept for the aspect of motivation that considers what people believe to be *worth pursuing* (Liu et al., 2022; Schwartz, 2012). Value systems are complex and dynamic belief structures that are informed by individual experiences and heavily affected by cultural settings. As individuals are involved in numerous cultures (Goffman, 1956; Hofstede, 1980) based on different forms of life (Cohen et al., 2001; Saariluoma et al., 2016; Wittgenstein, 1953), the weightings that they assign to different values are likely to change when moving between those different forms of life (Liu et al., 2022; Schwartz, 2012). Thus, an individual may interpret a similar type of eliciting event based on a different value system when the context has changed[46] or even in the same context but at a later point in time. Finally, the level of discourse links the individual level of moral thinking to an explicit social refinement of the kind of conventions that are formed and how they are connected to shared ideas of *something worth valuing* (Habermas, 1995; Nussbaum, 2004). Another way to state this is that social conventions are refined to reflect consensus on value priorities and hierarchies.[47] As such, the component processing model addresses discourse more implicitly than explicitly by considering how the appraisal of an event is congruent with ideas of good that are partly external to the individual.

The differences between considering the connection of emotional information processing and moral thinking through the lenses of *valuing* or *categorical moral emotions* requires further elaboration. The main difference is in the level of emotional information processing that is appealed to. Valuing is considered from the level of involuntary appraisal patterns related to micro-valence (Haidt, 2001; Shuman et al., 2013) or on the dimensional level of conscious feeling in the form of macro-valence attributed to an event (de Sousa, 2022; Wallach et al., 2010). In either case, valence is considered the main means of uncovering *individually meaningful* dimensions (i.e., the attribution of value) of a situation or *recognizing normative* aspects of a situation, event, or object (Haidt, 2001; Malle et al., 2020, 2021; Malle & Scheutz, 2019). Meanwhile, the conceptual level of categorical moral emotions is best described in terms of emotion themes that connect to ideas of normativity or harm: when self-directed, they have no meaning without a relation to a normative social sphere (Gomila & Amengual, 2009); when other-directed, they have a connection to an interpretation of a personal offence (P.S. Russell & Giner-Sorolla, 2011). Like basic emotions, categorical moral emotions have not been established by consensus, though suggestions include disgust, pride, shame, guilt, elevation, gratitude, awe, sympathy, empathy,[48] loyalty, anger, indignation, and contempt (Haidt, 2003; Laakasuo et al., 2021; Nussbaum, 2004; P.S. Russell & Giner-Sorolla, 2011; Tangney et al., 2007; Power & Dalgleish, 2016).

In the next section, we discuss how the connection between automated emotion inference technologies and approaches of moral sentimentalism can provide possibilities for building moral competence in AI artefacts. However, before presenting application suggestions, we examine whether the foundational issue of *Hume's guillotine* proves to be an insurmountable obstacle for such objectives. We first present the argument that Hume's guillotine is itself based on false assumptions and that correcting such assumptions eliminates the obstacle. Subsequently, we present and analyze views from contemporary literature.

3.1. Deriving values from facts

The role of facts in generating moral competence in AI artefacts has been a critical problem in innovating and designing concrete solutions for morally competent artefacts in general (Anderson &

Anderson, 2011; Müller, 2020; Wachter et al., 2021; Wallach & Allen, 2009). The main issue is whether it is possible to derive values from facts in the first place. According to Hume (1972), "there is no ought from is" (pp. 241–250). This aporia is called Hume's guillotine, or the *is–ought* problem (Pigden, 2010). We argue that hybrid forms of moral sentimentalism have an important role in solving this problem.

Hume's guillotine is based on the cognitive research of his time. In developing the construct, Hume first contrasted reason and emotions to argue—erroneously, from a modern perspective—that reason (i.e., cognition) cannot impact emotions, and it is thus impossible for it to affect moral sentiments. Considering modern research, it is evident that the human body (including but not limited to the brain) and mind operate as a whole, and the relation of emotions and cognitions could be likened to the relation of color and outline in vision—as two separated entities. Each situation is encoded by cognitions and emotions (Churchland et al., 1994; Holtzman, 2018).

Another flaw in Hume's argumentation is his underestimation of the meaning of actions within human thinking. People who act in the world inevitably end up in different situations. People automatically associate these situations with more or less positive or negative valuings. In this way, situations evoke mental states with different emotional valences. Actions that lead to negative situations are mostly interpreted as something to avoid, whereas actions that lead to positive situations are easily interpreted as aspirational. Over the course of their lives, people adopt private subconscious and conscious rules and practices that guide their actions (e.g., Bandura, 1991). As they collaborate with others, their personal rules are put in discursive relation to the rules of other people (Habermas, 1981, 1995). From family breakfasts to parliamentary debates, social discourses generate culture-specific moral systems. From Westermarck's (1906) perspective, such discourses create moral rules for cultures. In this dynamic social process, the human mind combines facts, actions, and ethics into information about what should be done, what is forbidden, and what is allowed (Saariluoma, 2021; Westermarck, 1906). For example, information about the lethality of smoking made the habit of smoking ethically doubtful. The statistical information showing a connection between illnesses and smoking and the painful nature of those illnesses created largely accepted anti-smoking ethics.

Explicating the false presumptions in Hume's thinking shows how the attribution of emotional information to the outcomes of actions enables people to circumvent Hume's guillotine. The combination of automated emotion inference artefacts and moral sentimentalist approaches promotes the same means of building a *flexible* form of moral competence (not merely rule-following) in AI artefacts. There are two stances on the use of information and facts to construct moral competence. First, people can give moral criteria to machines that classify cases as morally acceptable or unacceptable. For example, a machine may decide who will get a vaccination when the vaccine is administered only to people older than 65. Artefacts with this kind of moral competence can be called weak ethical AI (Saariluoma, 2021). It is also possible to ask if data masses can be used in the process of deriving new moral rules and values. With action information (i.e., representations of the end results of possible actions and the valences attributed to them), situation analysis, and facts, it seems theoretically feasible to create strong ethical AI, which are AI artefacts that can be involved in constructing moral ideals (Saariluoma, 2021). Thus, hybrid forms of moral sentimentalism illustrate the false assumptions underlying Hume's guillotine and prove to be important for the design of moral competence in AI artefacts by means of automated emotion inference technologies.

3.2. Competence as moral sensitivity and high-level moral regulation

Table 1 provides an overview of contemporary suggestions in the literature regarding the use of automated emotion inference in AI ethics. As the approaches are considered for building some form of moral competence in AI artefacts, the design suggestions require an explicated connection between the emotional information processing and morally relevant information processing of the person under evaluation (and/or of the artefact itself). In Table 1, this explication can be seen in the combination of the *Goal* of the emotion task and the *Role of AER* constructed from each reference. Together, they reflect the type of *moral competence* that the automated emotion inference technology is expected to support. *Role of AER* specifies the role of automated emotion inference in the tasks.

Table 1 shows how the referenced literature differs in the abstractness of its suggested applications. Some authors are more concerned with abstract goals and emotion tasks that concrete applications should consider (Coeckelbergh, 2010; Holst, 2021; Maldonato & Valerio, 2018; Nallur & Finlay, 2023; Wallach et al., 2010), some describe fairly abstract goals and relatively concrete situations, which further limits the dimensions for concrete applications (Balkenius et al., 2016; Crnkovic & Çürüklü, 2012; Gomila & Amengual, 2009; Liu, 2021), and others describe concrete applications that they link to a (seemingly) well-framed moral competence (Gloor, 2022; Tzeng, 2022). The idea behind the most abstract views is to *describe and argue for* the importance of considering emotions when designing forms of moral competence in AI artefacts. Authors who make use of fairly abstract goals (e.g., avoidance of risks and danger) and describe fairly concrete task situations (e.g., clearly explicating the interaction situation as the context for attaining the goal) go a step further by *suggesting aspects* of the interaction that are relevant for concrete applications (e.g., the presence of categorical moral emotions), while noticing that scientific validation of the suggestions is still required. Authors who describe concrete applications *not only suggest but also select* aspects of an interaction situation to produce a framed moral competence in the resulting artefact (e.g., considering categorical emotional responses to a controlled eliciting event as indicative of the values a person considers important (Gloor, 2022), or expressions as indicative of an action being congruent or incongruent with an implicit social norm (Tzeng, 2022)). In such cases authors consider the selections to be scientifically validated.

Importantly, the abstractness of the suggested application does not necessarily affect how concrete levels of emotional information processing are appealed to when explicating the role of automated emotion inference within the task. For example, Wallach et al. (2010) identify the highly abstract goal of modeling moral reasoning capabilities to artificial general intelligence but the fairly concrete idea of automated emotion inference as a means of *valence tagging*. In contrast, Gloor (2022) describes concrete applications of image recognition techniques that compare information from facial action units to learned categorizations of emotion responses and select *the values* of the person by comparing the inferred emotional expression to the controlled eliciting event (video material). Even if the technological considerations in Gloor's suggestion are concrete, the underlying models rely on a highly abstract understanding of emotional information processing and its connections to moral dimensions of thinking. The same is true in Tzeng (2022), who describes concrete methods for multi-agent modeling but does not address the requirements for further explicating the kind of information about emotional information processing that people's expressions convey or how the expressions could imply congruence or incongruence with norms that are represented in the models of a single artificial agent but not visible to others.[49]

We raise this tension between the suggested technical means and level of emotional information processing appealed to, to show how, within contemporary literature, authors who concentrate on concrete applications do not notice the leaps over theoretical gaps they make in their suggestions. This oversight leads to inconsistencies in the idea of operationalizing moral competence with the help of automated emotion inference in a similar manner as in automated emotion inference development in general. Failing to notice these gaps can easily lead to hubris when describing the purposes of the artefacts:

> Artificial Intelligence and machine learning give us a novel way to measure the true personality characteristics, morals, and values of a person, based on that person's behavior, their honest signals. These honest signals are the little words, the body language, the tone of the voice, the glances, the body posture, that a person shows in response to external events. (Gloor, 2022, p. 11)

Table 1 also reveals that many of the authors consider the role of automated emotion inference as a minimum requirement for *moral sensitivity* (i.e., the ability to detect morally relevant aspects of an event) of an AI artefact. This consideration is apparent from how most of the authors describe information gathered with the help of automated emotion inference to provide input for synthetic emotion architectures built into the AI artefacts. As Gomila and Amengual (2009) aptly summarize, "not just human emotion recognition in robots, but interaction in emotional terms, is required" (p. 166). This notion connects the idea of automated emotion inference and computational modeling of emotions as a means for high-level moral competence for AI artefacts.[50]

Table 1. Summary of approaches in the referenced literature.

Article	Suggested moral competence
Nallur and Finlay (2023)	**Goal**: To build an empathy capability in AI artefacts **Role of AER**: To "understand the emotional projection by the human," which is further used to "reason about the human's mental landscape, and respond accordingly" (p. 973)
Gloor (2022)	**Goal**: To recognize the values of individual people by inferring expressed emotions **Role of AER**: To interpret a categorical emotional response (from a facial expression) to a controlled eliciting event (video) to ascertain a person's values
Tzeng (2022)	**Goal**: To develop AI artefacts capable of considering human values and motivations and adapting to changes in norm structures within their environment by inferring emotional expressions to executed actions **Role of AER**: To transform the emotional expressions of others (both humans and artificial agents) into information about the desirability of a norm violation to lead to enforcing or inhibiting such violation (by the artefact) in the future
Liu (2021)	**Goal**: To develop social robots capable of recognizing people's emotions and regulating their own actions with the help of moral *ersatz* (synthetic) emotions **Role of AER**: "When designing affective robots, in addition to creating machine perception and computations that can recognize and understand human emotions, our focus is to build 'artificial moral sentiments' on the basis of artificial intelligence, to help the robot itself make ethically correct decisions" (p. 236)
Holst (2021)	**Goal**: To sketch out how emotions are important for rational and moral behavior and, as such, how they should be implemented in our vision of morally behaving artificial general intelligence **Role of AER**: To recognize emotions in people, as the capability to recognize emotions is a prerequisite for regulating adequate emotional (e.g., care) responses of AI artefacts
Maldonato and Valerio (2018)	**Goal**: To build intrinsic morality in AI artefacts **Role of AER**: To control the actions of AI artefacts by detecting the emotional states of others and reflecting action possibilities through *synthetic emotions* instead of mere logical reasoning
Balkenius et al. (2016)	**Goal**: To build intrinsic morality in AI artefacts by instantiating learning capabilities in them that are grounded in sensory-motor interaction with the environment and the emotional experiences (of the artefact) that emerge from the interaction **Role of AER**: To be part of the mechanism for learning social expectations and emotional effects (negatively-positively valenced) of behavior by observing people's emotional (physical) expressions
Crnkovic and Çürüklü (2012)	**Goal**: "inbuilt self-regulatory behavior that makes a robot notice and avoid risks and dangers and learn from experience" (p. 64) **Role of AER**: To "assess the effects of their [AI artefact] actions on sentient beings and act accordingly" (p. 68, brackets added) with synthetic emotions as a regulatory mechanism for future action selection (e.g., remorse, regret)
Coeckelbergh (2010)	**Goal**: To build the appearance of emotional capabilities in AI artefacts to generate a capacity for moral behavior **Role of AER**: To provide means for the artefact to perceive cues from the external world, including when interacting with people, so it can react accordingly to happenings by launching synthetic emotional reactions to them
Wallach et al. (2010)	**Goal**: To build a model of artificial general intelligence capable of moral reasoning **Role of AER**: To support *valence tagging* of sensory input, including people's expressions. Valence tagging further affects the guidance of attention and the selection and execution of actions. Valence tagging and the regulation mechanisms it provides input for, is, as a total, considered a feeling entity that instantiates values of the AI artefact
Gomila and Amengual (2009)	**Goal**: "building systems with a practical understanding of emotions and emotional interaction, in particular moral emotions" (p. 162) **Role of AER**: To infer about the emotional expressions of others and feed results of the inference forward to a synthetic moral emotion structure for further analysis. Together, automated emotion recognition and the moral emotion structure would count as the *moral sensitivity* and *motivational* system of AI artefacts

It is important to consider which input automated emotion inference is proposed to provide for further processing in the literature, since it reflects the level of emotional information processing that the suggestion appeals to: categorical emotions (Gloor, 2022; Liu, 2021), emotional expression or projection but no further explication of its indication for the emotional information processing of a person (Coeckelbergh, 2010; Crnkovic & Çürüklü, 2012; Gomila & Amengual, 2009; Holst, 2021; Maldonato & Valerio, 2018; Nallur & Finlay, 2023; Tzeng, 2022), and valence (Balkenius et al., 2016; Wallach et al., 2010). When reflecting on requirements for successfully developing concrete AI applications, viewpoints that do not specify the information to be conveyed by *expressions* or *projections* leave an important phase of building moral competence open to interpretation. At the same time, our current understanding of valid automated emotion inference does not validate an appeal to the thematic level of emotion categories. Only the suggestions of Balkenius et al. (2016) and Wallach et al. (2010), who appeal to the level of valence can be considered consistent with our current knowledge of the possible capabilities of automated emotion inference. Even then, there is the further question of the connection between the micro-valence aspects (intrinsic pleasantness and goal congruence) and the macro-level of valence as a feeling.

The last important insight from Table 1 is that most of the authors describe moral competence as the *quality of behavior of the artefact* in an interaction situation with a human being, thus presuming that the automated emotion inference is directed at inferring individual-level emotional information processing. An exception is Gloor (2022), who sees moral competence as the ability of the artefact to reveal latent aspects of an individual's moral thinking *to people*.[51] Tzeng (2022) also differs by not assigning importance to single interaction situations per se but grounding moral competence in the aggregated information gathered from a vast number of interaction situations that are used to inform the emergence of new norm structures.

4. Elaboration of limitations

In this chapter, we synthetize the findings from the earlier chapters as limitations stemming from inherently involved areas of reasoning when automated emotion inference is operationalized as a practical solution in AI ethics.

4.1. Limitations as explicated emotion task

Moral competence involves the idea of moral sensitivity (Graff, 2024) as it implies the capability of sensing morally relevant content within a decision situation. Some authors have highlighted how such sensitivity may be manually coded (and updated) in AI artefacts (IEEE, 2019; Malle et al., 2020) after people have recognized the relevant content for coding (e.g., social norms). Authors with a stricter interpretation of moral sensitivity have argued that coded sensitivity cannot be considered a *capability to sense* morally relevant content because there will always be occasions where a coded sensing mechanism will lead to contradicting decisions (Coeckelbergh, 2010), and it implies an overextended analogy between the capabilities of the artefacts and humans. We agree with this latter argument and propose that the idea of moral sensitivity in the context of moral competence of AI artefacts should be referred to as building "inference mechanisms" rather than "sensing capabilities" in the artefacts.

However, we contend that the main reason to use automated emotion inference as part of embedding ethical capabilities into AI artefacts (Salo-Pöntinen, 2021) is to provide the artefact with some level of flexible moral sensitivity. Flexibility in this context refers to the advantage, in comparison to rule-following sensitivity mechanisms, that automated emotion inference could provide for understanding the moral dimensions of events. Instead of focusing on fixed rules, the artefact would have an inference mechanism for discovering dimensions of events that should be considered within a single action selection or when designing good conduct. Therefore, the high-level emotion task of automated emotion inference in AI ethics is being part of a moral sensitivity mechanism of a technological system. This high-level task sets the first limitations for understanding automated emotion inference as a competence mechanism in AI ethics.

Following the referenced literature, moral sensitivity can be divided into two dimensions: the inference of morally relevant mental content of people or the inference of emotional content that is relevant for moral interaction. The latter is linked with the idea of providing higher-level moral competence for the artefact itself and, as such, requires the explication of regulation mechanisms that react to the feed provided by the automated emotion inference mechanism. Suggestions for such regulation mechanisms include synthetic emotions, which require input from a closely linked field of computational modeling of emotions. In the simplest terms, synthetic emotions consider the inhibition or reinforcement of an action selection according to the inferred emotional responses of the other parties in the situation. Discussions of the development of synthetic emotions are still in their infancy (Battaglino, 2013; Battaglino & Damiano, 2015; Keltner et al., 2023; Ohja et al., 2021). Thus, ideas for using automated emotion inference to build high-level moral competence in AI artefacts are also still highly theoretical. Further limitations set by the concrete emotion task(s) of a design must be analyzed in the design process as described in Section 2.1.1 (*The emotion task and inherent areas of reasoning*).

4.2. Limitations of emotion studies and moral sentimentalism

This section considers the limitations of *what can be inferred* in the first place about the emotional information processing and morally relevant mental content of people. As noted, there are four conceptual levels for understanding the kind of morally relevant content that can be inferred from people's emotional information processing and the level considered as the linking phenomenon: valuing (valence), complex moral nuances indicated by categorical moral emotions (categorical emotions), values of individuals (emotional attitudes), and value refinement (valuings explicated in discourse). From these levels, our current knowledge on scientifically valid automated emotion inference limits the possibilities to make inferences about specific aspects of micro-valence (goal congruence, intrinsic pleasantness) when examining overt expressions and to evaluate emotional attitudes on the population level by considering text-based automated emotion inference. Therefore, our current knowledge limits overt approaches on the valuing level as well as text-based approaches for analyzing valuings explicated in discourses but not for analyzing moral content on the individual level. Notably, even when attitudes are simplified as an emotionally valenced disposition toward some entity (Giner-Sorolla, 1999; Oatley et al., 2006), the dynamic nature and context specificity of the values of an individual *undermine any attempts to recognize the values of individuals* by means of automated emotion inference.

For overt approaches, there are further limits to what can be inferred from indications of the micro-valences related to goal congruence and implicit pleasantness, as inferences are always based on the expressions of individuals. In the emotional information processing of an individual, the underlying latent information regarding *what the eliciting event is* (e.g., it can be purely a memory without reference to the physical or social context), *what the interpreted social context is*, and *whether the elicitation is intentional or not* (i.e., acting or not) is always present. Thus, even in the best-case scenario, it is at most possible to infer that something elicited the micro-valence appraisals and that they provide non-directional hints about the valuings of the person. Further analysis—for instance, to identify the subject of appraisal and the consistency of the indicated valuing (i.e., whether all five micro-valence aspects are congruent) requires the explication of the person's interpretations of the event. Therefore, design ideas that rely on understanding the target of valuing or on more nuanced meaning attributions involved in the situation must invite individuals to communicate their interpretations.[52] However, this is not always possible, as individuals must be able to categorize and verbally express this information in order to communicate it. Finally, given the contemporary view that automated emotion inference recognizes *the values* of a person or conveys *the emotional profile*, we emphasize that the inferences produced *cannot be understood as stating anything about the person exterior to the single event or context*. Emotional information processing and morally relevant mental content are highly temporal and context dependent.

The findings within emotion studies and moral psychology give only theoretical insights for constructing valid overt automated emotion inference. Therefore, such findings can *in no manner* be straightforwardly converted into part of automated emotion inference artefacts, which has been the mode of action in contemporary approaches. More experiments are required to confirm the findings, and a vast amount of field experiments should be conducted to test the ecological validity of possible automated emotion inference pursuits that rely on laboratory findings. However, since it is likely that eliciting events are controlled even in field experiments, such experiments cannot definitively determine *what* the eliciting event is in a given moment, which is an ever-present question when using the third-person view to make inferences about emotional information processing in real-life settings.

Because of these limitations, the appropriate use cases of automated emotion inference mostly target the recognition of phenomenon on the group level with aggregated information from well-defined populations. In most situations, they are not suitable for application on the individual level. There are a few cases of well-defined and highly beneficial situations (for the individual, not the deployer) in which models relying on aggregated population information could arguably be useful for making inferences about an emotional phenomenon on the individual level. Such situations are highly rare, however, as the costs for individuals due to the impreciseness of the artefacts far outweigh the benefits in most cases. Some exceptional cases include using the models to help recognize indicators of pain in patients who are unable to communicate about it themselves, therapeutic applications for helping people learn

emotion regulation skills, or alerting drivers to indications of fatigue (Bondioli et al., 2024; De Prisco et al., 2023; Gratch et al., 2024). Still, even in such cases, the use of the models is not problem free.

4.3. Technical limitations

In this section, we discuss the technical limitations of overt automated emotion inference in the form of detecting micro-expressions, as this modality has yielded the most promising results. For other modalities, such as detecting tone of voice (Scherer & Moors, 2019), or micro body gestures (Chen et al., 2023), their particular technical limitations must be considered accordingly. Mohammad (2022) has addressed the technical limitations of text-based approaches, but it should be noted that the novel uses of large language models for enriching text-based automated emotion inference poses novel questions (Yang et al., 2023). Before turning to the case of detecting micro-expressions, we highlight that in addition to having modality specific limitations, most automated emotion inference technologies are based on statistical machine learning models. Thus, they are prone to issues such as encountering novel scenarios (i.e., out of distribution problem (Yang et al., 2024)) or data (i.e., domain shifting (Zhang et al., 2021)), that have not been present in their training data, or to black-box and related explainability difficulties (Koshiyama et al., 2024). These general issues emphasize the importance of evaluating limitations for what the produced models can be considered to infer, not merely from emotion research and moral sentimental perspectives, but also from limitations stemming from the statistical foundations of the used models.

The detection of micro-expressions is not an easy task (Varanka et al., 2024; Zhang & Arandjelović, 2021). In psychological experiments, their detection has been based on observation experiments examining facial action units (Ekman & Friesen, 1971, 1978). While these action units were initially believed to provide information about discrete emotions, contemporary experiments have considered them as part of involuntary physical responses to micro-valence appraisals. The range of useful action units has also been narrowed down to areas involved in involuntary responses, namely the zygomaticus (cheek) and corrugator (brow) areas, where the steady reactions (i.e., muscle activations) are so minute and rapid that they have been detected in contemporary experiments using electromyography technologies instead of visual observation (Gentsch et al., 2020).

Therefore, a technological limitation is that we might be able to recognize steady hints of goal congruence and implicit pleasantness in a laboratory setting, but the possibility of reliably detecting them in real-life situations through physically[53] non-invasive mechanisms, such as video recognition, still remains an open issue. Contemporary automated emotion inference pursuits for non-invasive micro-expression detection (e.g., Ben et al., 2022) may provide grounds for development in the form of being part of proving that micro-expression detection from visual material is technically possible. However, since contemporary pursuits are heavily founded on psychologically invalid presumptions, we do not consider them sources of evidence of reliable micro-expression detection *relevant for automated emotion inference*. In this sense, if enriched with expertise in emotion research, studies that do not claim to detect emotions but merely consider the technical boundaries of detecting micro-expressions from visual material, may prove to be a better offset for future research (Varanka et al., 2024).

4.4. Ethical limitations

Automated emotion inference is always invasive in a psychological sense, as it aims to recognize hints about a person's internal world without the person actively enclosing them (Corvite et al., 2023; Howell et al., 2018; McStay, 2020; Nakamura, 2019; Ullstein et al., 2022). This invasiveness is one reason why automated emotion inference should not be pursued on weak grounds and definitely not in delicate settings, such as evaluation contexts characterized by power imbalances (e.g., hiring situations, performance evaluations in work or education). Such invasiveness is less pronounced when the technology provides information about populations of people rather than about individuals. However, this point of view is often mistaken as analogical to the privacy concern of not identifying a person's identity when automated emotion inference technology is used (Uddin et al., 2024; see also Boutin et al., 2023; Fu & Lyu, 2025). As we have pointed out, invasiveness is already present in the claim of being capable of

detecting the emotional information processing of a person, regardless of whether the person is identified at the same time (McStay, 2020). This point is emphasized by the notion that even an artefact that does not perform identification still produces inference results that have (often considerable) effects on the individual (Howell et al., 2018; McStay, 2020; Monteith et al., 2022; Nakamura, 2019; Roemmich et al., 2023; Stark, 2023). It should also be understood that even population-level use is still a use of technology to make inferences about the latent mental content of a group.

It has often been argued that the application of automated emotion inference facilitates fairer treatment of individuals since the alternative, human evaluation, consists of biased individual judgements (Gloor, 2022; Roemmich et al., 2023). Such statements are misleading and ignore the fact that third-person evaluations conducted by individuals and by AI artefacts are significantly distinct activities, and assessments of their validity and desirability should thus rely on distinct criteria. It is true that different forms of bias affect people's judgements about other people's characteristics (Gorman, 2005; Todorov et al., 2015). However, the potential negative effects of using inherently biased automated emotion inference to evaluate individuals are overwhelmingly greater than those resulting from individually dependent biases. The day-to-day emotion attribution of individuals rarely impacts large groups of people,[54] but the models that are adopted always do (Ajunwa, 2022; Boyd & Andalibi, 2023; McStay, 2020; McStay & Rosner 2021; Roemmich et al., 2023; Stark & Hoey, 2021; Stark & Hutson, 2022). Therefore, the argument that contemporary automated emotion inference is *more objective* than individual people essentially says: at least an aggregated model has the same biases toward all candidates.[55] In extreme cases, the models may even be taken as truth and lead to a *locked position* that is not reflected upon (McStay, 2020; Robert et al., 2020; Stark, 2023). In addition, calls for the objectivity of AI applications do not recognize that all AI applications are based on numerous operationalization procedures that are affected by the people involved (Birhane, 2021; Gitelman, 2013; Jasanoff, 2016; Shilton, 2018; Zicari et al., 2021). This aspect undermines any claim of initial or inherent objectivity of the technology and highlights the need to understand the discourses underlying the operationalization procedures to permit an evaluation of the degree of the application's objectivity within the task context. Without such evaluation, arguments have no explicit and valid referent for objectivity (Ajunwa, 2020; Bernstein, 1983; Gitelman, 2013).

There is also a substantial difference in the flexibility and potential impact of individual day-to-day decisions compared to the formation of large AI models that utilize aggregated population-level data to evaluate large groups of individuals. Individuals are somewhat flexible in their day-to-day thinking, as they can detect mistakes and correct misconceptions, while models do not have a similar kind of flexibility.[56] People make day-to day decisions to be capable of acting within their (social) environments, whereas the use of automated emotion inference is often based on economic interests, such as increasing the attractiveness of a product to customers or making people more effective workers. To balance economic interests with, for example, social sustainability, one must reflect on the possible effects of building and using the model *that extend beyond the immediate use situation* (Mohammad, 2022; Roemmich et al., 2023; Sigfrids et al., 2023). Finally, the use of automated emotion inference applications constitutes a kind of *intervention* in the social context where they are deployed. As such, their influence on the context differs from that of person-to-person attribution (Howell et al., 2018; Nakamura, 2019). Their uncharted effects may undermine the positive goals of interventions, such as enhanced employee wellbeing.

4.5. Appropriate development and use framed by concrete emotion tasks and the combination of limitations

As noted in Section 2.1.1, the last area of reasoning inherently involved in all development and deployment of automated emotion inference in AI ethics is the combination of limitations to frame an evaluation of the *appropriate use* of the resulting technology. Here, our main purpose is to emphasize the existence, difficulty, and meaning of this last area of reasoning rather than to examine it exhaustively.

The insights we have provided about the difficulties with operationalization processes of automated emotion inference are highly important because they reveal the complex point of junction that the design of intelligent technologies cannot overtake. Different research fields have their own research

paradigms to frame relevant research questions and objects of research, theoretical concepts, and research goals (Kuhn 1996; Saariluoma, 1997). For example, engineering seeks to produce concrete instantiations of abstract ideas (Jones & Gregor, 2007; Lee et al., 2023; Simon, 1970). In AI development, this pursuit imposes the requirement of formalizing the ideas into a computable form (Fish & Stark, 2021; Russell & Norvig, 2016; Saariluoma & Karvonen, 2024), which is often paired with a generalizability requirement of the resulting models (Cockton, 2004; Prabowo et al., 2023). Within emotion studies and moral sentimentalism, the main aim is to understand the research phenomena (Barrett et al., 2019; Cowen & Keltner, 2017; Kauppinen, 2022; Power & Dalgleish, 2016), which may yield prediction capabilities in controlled settings (Scherer, 2009; Scherer & Moors, 2019) but often also sets limits for what can be generalized since such phenomena are known to be highly correlative to ontogenetic development and contextual influence (Liu et al., 2022; Power & Dalgleish, 2016).

When a phenomenon mainly studied by means of natural language is aimed to be communicated in a computable form, the transcription (i.e., formation of generalizations of concepts and their *relevant* connections that *frame* the phenomenon) (Khare et al., 2024; Prabowo et al., 2023) always requires an approximation of information in which something is left out to enable transcription. Therefore, even if the different areas of reasoning related to operationalization are not explicitly understood, presumptions made in them are implicitly included in the produced artefacts and may be evaluated as valid or invalid. Therefore, operationalization is the point at which a critical understanding of the justified use contexts, tasks, and limits of the produced artefacts can be formed. Such understanding should be the basis for recognizing the artefact as valid for its purpose (Branford, 2024; Mohammad, 2022).[57] Thus, the nature of this area of reasoning is best illustrated as an understanding of the development of concrete artefacts as an *instantiation of approximated information from relevant fields*. The general aim within this area of reasoning is to guide the selection of ideas that would be instantiated according to their desirability and ethical limitations and to comprehend the effects of approximation requirements to appropriately evaluate the outcomes of the produced artefact.

Here, we elaborate an example case using limitations mentioned in this chapter to demonstrate this last area of reasoning. Because of the psychological invasiveness and inherent impreciseness of using automated emotion inference to make inferences about individuals, it is not easy to find cases that are ethically sustainable (notwithstanding the few well-argued examples). Therefore, we suggest that, when used, such technology should first and foremost target an understanding of population-level phenomena. The use and purpose of the technology should be openly communicated to the people who may be exposed to it, and they should have the opportunity to reject that exposure. As the micro-valence aspects of implicit pleasantness and goal congruence can be considered indicators of non-directional valuings of people in a given situation and possibly captured by image recognition, we propose that a concrete emotion task could be the detection of situations where people seem to exhibit low or high valuings, as such cases can be relevant for further moral evaluation. Before concrete applications, the ecological validity and possible yet undetected limitations of such use of the laboratory findings should be examined in field experiments. Furthermore, if contemporary technical limitations are overcome, the automatized aspect of the task can involve the use of image recognition techniques to infer indications of micro-valence that appear in the form of relevant activations of action units in the brow and cheek areas of people. Once situations with low or high valuings are indicated by the artefact, the situations and possible valuings can be better understood by inviting people to communicate their interpretations of the situations. Further analysis may show that, in reality, the *flagged* situations (i.e., those marked as important by the artefact) involve too many instances of differently directed valuings to be useful. In fortunate cases, it may provide a more nuanced understanding of the moral dimensions related to the situation by making it explicit what is most often the valued or devalued entity and why. This knowledge could support a better understanding of desirable events in the situation or, for example, an explicit discussion of whether a certain intervention should be implemented to avoid unvalued events.

This example reflects how, even though people are unconsciously guided by their affective responses, norms, and value structures, making these factors explicit can allow people to reflect on their collective effects to their belief structures, which further enables to reform said belief structures (Nussbaum, 2004). In this regard, one purposeful use of automated emotion inference in the context of AI ethics

could be to develop a moral sensitivity tool that can illuminate events and cases for further reflection and discourse within a community. Such use emphasizes meaningful *interaction* that an interface may support in a similar manner that Boehner et al. (2007) consider an appropriate goal for the design and use of automated emotion inference in general: "Measures of success for such [interaction oriented] systems therefore do not focus on whether the systems themselves deduce the 'right' emotion but whether the systems encourage awareness of and reflection on emotions in users individually and collectively" (p. 276, brackets added). Use aiming directly to disclose *the values* or *valuings* of individual people or groups of people is, scientifically and ethically speaking, always problematic.

5. Conclusions

In this article, we sought to answer two questions: (1) Are the attempts to design moral competence in AI artefacts by means of automated emotion inference techniques relevant? (2) If so, what kinds of limits should be acknowledged? We have answered the first question in two parts. First, we found that much of the contemporary approaches of automated emotion inference are scientifically biased. Prevailing presumptions have led developers and deployers to believe that emotional information processing does not need to be considered in the conceptual landscape of intentionality because the discreteness and temporality presumptions offer a path through such latent phenomenon. In this way, contemporary automated emotion inference has mistaken the epistemological commitment of basic emotion theories (i.e., we *can gain meaningful information* about a limited set of emotions by observing people's expressions) for an ontological one (i.e., the observed expressions *are* a direct causal link to people's mental states, so one can hold expressions as emotions). These invalid presumptions are also evident in concrete application suggestions for using automated emotion inference to build moral competence in AI artefacts.

Even if current views are not scientifically speaking relevant, changing the focus of automated emotion inference from behavior to action reveals paths to valid design possibilities. Such shift requires the acceptance of intentionality as a concept that cannot be dismissed when considering the emotional information processing of people. To acknowledge intentionality, one must observe that there are always multiple forms of latent information processing involved in the encoding of emotions and even in the seemingly straightforward phenomenon of natural emotion recognition. Mental representations, apperception, and appraisal are crucial concepts for acknowledging and analyzing the latent phenomenon. Experiments relying on those concepts have produced sufficient knowledge of involuntary mechanisms to, on a theoretical level, implement them in automated emotion inference technologies. However, even in the best-case scenario, automated emotion inference can only provide hints about the emotional information processing of a person. It is important to note that difficulties will remain in the detection of the eliciting event and the latent social context involved even if overt automated emotion inference is developed according to promising findings on micro-valence. When considering the applicability of the findings from a moral sentimentalism point of view, one can at most infer that something elicited the micro-valence appraisals and that these appraisals represent non-directional hints about the valuings of the person. For further analysis—for instance, to determine the subject of the appraisal and whether the indicated valuings are consistent (i.e., all five micro-valence aspects are congruent)—it is necessary to invite the individuals to explicate their interpretations of the event. Based on these findings, we conclude that a heavy change in focus and goals could grant relevance to attempts to design moral competence in AI artefacts by means of automated emotion inference.

Our main finding for the question of what kinds of limits should be acknowledged is that, within operationalization processes aiming to develop concrete instantiations, limits should be acknowledged according to the concrete emotion task and the combination of limitations set by inherently involved areas of reasoning. Such framing recognizes the difficulties of developing increasingly sophisticated technologies. In particular, the operationalization of the concepts and phenomena is the point at which the good intentions of designers can easily go astray, as designers may fail to acknowledge the limits of their own expertise.

Since our article considers methodological insights, we can discuss limitations only on a high level. The first limitations derive from the high-level emotion task of using automated emotion inference in

AI ethics as part of the moral sensitivity mechanism of a technological system to achieve sensitivity to *morally relevant mental content* of people or emotional responses that are relevant to *moral interaction*. The inherent areas of reasoning that pose further limitations are described as follows: what can be inferred about the emotional information processing of people in the first place and how it is connected to moral thinking (limitations from emotion studies and moral sentimentalism); what the technical requirements are (technical limitations); what kind of technology is desirable (ethical limitations); and which development and use is appropriate as framed by the concrete emotion task and the combination of limitations (understanding instantiation as the approximation of views from different fields). From a design point of view, contemporary automated emotion inference is an example of technology-driven design processes that lead to an oversimplification of the areas of reasoning related to human dimensions of technology design. As all technology involves designing human-technology interaction, the technology driven processes commonly produce insufficient products from aspects such as usability, likability and placing in people's lives. However, as we have shown, in cases where the technology use targets phenomenon that fall in the area of human research, technology driven processes produce straightforwardly invalid products.

It is easy to trivialize the richness of our moral thinking. Thus, our main goal has been to guide the theoretical and ecological validity of attempts to embed ethics in AI artefacts through automated emotion inference techniques. Altogether, valid treatment of such pursuits entails abandoning the thought of recognizing any mental content of people as such. In most of the cases we reviewed, acknowledgement of the validity requirements meant abandoning the original goals. However, when emotional information processing and its connections to moral thinking are understood as a dynamic, multicomponent, and multi-level appraisal process, we can operationalize well-defined and empirically valid emotion tasks that may be (partially) automized to gain a better understanding of what people consider desirable or undesirable in a given situation. This framing is (justifiably) far from the promises made by contemporary literature and applications, but it is a way in which automated emotion inference can enrich the moral discourses taking place in communities (i.e., contain a form of moral competence).

Despite its limitations, this type of moral competence is a flexible and dynamic one. It is involved in the process of *constructing* moral ideals within communities and not merely a *manifestation of ideals* in the form of formalized rules. Our argument is that hybrid forms of moral sentimentalism open a path to solving the aporia of Hume's guillotine. They reveal processes related to the human ethical cycle: the attribution of valuings to the end results of actions—the construction of subconscious and conscious rules based on the valued and unvalued actions—the assessment of individual rules in discourse—and the construction and explication of moral ideals through discourse. It is possible to simulate this cycle and use AI artefacts on this ground in the process of generating ethical rules. In this way, the solution to Hume's guillotine opens up possibilities to create forms of strong ethical AI.

We have concentrated on the foundational issues of using automated emotion inference as a practical solution in AI ethics. Our purpose has been to clarify the existence of such a problem space and provide correcting views when possible. Future research could benefit the methodological discourse by further analyzing, for example, the relations of indicators of implicit pleasantness and goal congruence to the macro-level valence. Such research benefits from considering hybrid approaches to moral sentimentalism that focus on the integration of cognitive-affective processes as means of understanding moral thinking. Furthermore, Following Boyd and Andalibi (2023) and Roemmich et al. (2023), the findings of this article could be combined with a values lens method to systematically uncover contradictions between the organizational problems that contemporary autonomous emotion inference technologies supposedly solve, and the problems involved in the suggested solutions. Such research could yield important insights for discourse on the desirability of using automated emotion inference to build moral competence in AI artefacts.

Notes

1. With automated emotion recognition, we refer to technical artefacts that are built for the purpose of sensing and interpreting (sometimes combined with the term "detecting") people's emotional information processing. The technical means used to build such capabilities vary hugely, but a major distinction can be made

between approaches that use textual information and approaches that use overt physical indicators as the input for further interpretation (Mohammad, 2022; Picard, 1997).

2. Moral competence recognizes that designing AI artefacts to have full moral agency is not feasible, but we can design AI artefacts that can sensibly be considered morally capable in a narrow context and manner (i.e., to have some kind of moral competence) (Malle & Scheutz, 2019).
3. Following Scherer and Moors (2019) (see also Scherer, 2009), we use "emotional information processing" interchangeably with "affective information processing." We understand emotional phenomena as multi-component, multi-level appraisal processes that are elicited by the interactions people have with their environments (internal reality, i.e., memory, and external reality). We elaborate on emotional information processing in the following subsection.
4. See Salo-Pöntinen (2021) for further discussion on practical solutions within AI ethics.
5. The research method is called "foundational analysis" (Saariluoma, 1997), which we detail in the following subsection.
6. "Valence" refers to experiencing a feeling as positive, neutral, or negative and to the underlying micro-valence appraisals (Russell, 2003; Shuman et al., 2013).
7. Here, non-personalized means that the analysis is conducted on the level of general language use, not on the level of what individuals have pursued to convey with their language use. See Section 2.1 for further elaboration on this highly important nuance.
8. The primary goal of automated emotion recognition should be to recognize population-level phenomenon. The inherent impreciseness of recognizing individual phenomena imposes serious ethical and validity boundaries on its use to make inferences about individuals. We elaborate on this aspect in Section 2.1 and further discuss its implications in Section 4.2.
9. The concept of an emotion task was constructed for automated emotion recognition design to make it clear from the outset of a design process that a third-person evaluation cannot fully recognize the mental state of a person but that it *might* be possible to recognize some aspect of a person's emotional experience at a single point within a clearly articulated context (i.e., a dimensionally limited state) (Mohammad, 2022). Therefore, analysis of the use situation (and explication of a task framed accordingly) is a main instrument for appropriately designing and deploying automated emotion recognition.
10. In other words, the level of field relevant scientific expertise required for the operationalization of concepts within AI design and development should be similar to the level of field relevant scientific expertise required to use those concepts in scientific experiments. We parallel this requirement with theory building since the design-oriented literature has widely accepted that the theories supporting concepts in AI development must be scientifically valid. Therefore, with our formatting of the problem, we insist on expanding an already recognized validity statement.
11. Moral sensitivity is the capability to detect morally relevant aspects of an event (Graff, 2024).
12. For an exception, see Mohammad (2022) and Branford (2024), who uses the term "affect recognition technologies" (ART), as well as Katirai (2024), who uses the term "emotion recognition technologies" (ERT).
13. There are multiple proposals for HCI research to get more involved in issues related to artificial intelligence (Harper, 2019; Ozmen Garibay et al., 2023; Shilton, 2018; Stephanidis, et al., 2025) and affective computing (Boehner et al., 2007; Gaver, 2009). The use of automated emotion recognition in AI ethics requires to frame ways for HCI research to deal with them both.
14. Grammars of action are the implicit regularities within given forms of life that give meaning, and which can be used to construct new meanings for language use in language games (Wittgenstein, 1953). In Wittgenstein's late philosophy, grammars of action as the implicit regularities residing in forms of life is a postulate for meaningful language use.
15. Neal et al. (2019) have provided a substantive analysis of the use of psychometrics in general. The development of automated emotion recognition is based on automatizing psychometric practices.
16. For views that combine both aspects, see, for example, Katirai (2024).
17. The discussion presented in this section concerns single modality as well as multimodal approaches (Zhang et al., 2024).
18. Sentiment analysis is a textual analysis approach to automated emotion recognition (Mohammad, 2022).
19. See especially the part concerning *Digital Minds*.
20. Such reinforced learning takes self-reports of individuals as input for validation or correction. However, the self-report functions rely on the same operationalized presumptions that caused the mistakes in the first place. Thus, the *corrective self-report* is just another case of scientifically invalid forced choice.
21. It is sometimes implied that emotion regulation is equivalent to lying about one's *true* emotions (Gratch et al., 2024). This idea is not at all in line with emotion studies, which have considered emotion regulation as an integral part of the emotional information processing of individuals (Folkman & Lazarus, 1988; Gross, 2008).
22. The concept of "emotional state" may be useful in designing interaction-critical aspects of an artefact (Saariluoma et al., 2016). While it is a *good enough* concept for this purpose, its usage still offers a limited view of the emotional information processing of people. In addition, it does not presume to predict

individual people's mental states at a given moment but rather to enable the consideration of emotional information processing as part of the interaction design.

23. Focusing on one's preceding experiences can also evoke involuntary reactions and expressions related to the recalled memory. This possibility for elicitation and control has been used in psychological experiments (Scherer, 2009) and is well known in the field of performing arts (Benedetti, 2005).
24. Different forms of behaviorism were influenced by the (now-abandoned) theory of the scientific paradigm of logical positivism (Graham, 2023). As an analogy to our findings, one can find concealed logical positivistic practices from other fields relevant to technology development (Siponen & Klaavuniemi, 2021).
25. These have been present in the work of Ekman (1993) as well, though not in a fine-tuned version. Ekman's work discusses emotion families and their construction and differentiation within continuous dimensions that are present in affective states.
26. Guerrero et al. (2023) have offered a good example of how already acknowledging the limits of a model lessens the risk of overstating its appropriate use. However, even these researchers rely heavily on false presumptions and, as such, risk disseminating a false validation of those presumptions to be used in future research as well (Mertala et al., 2022). To eliminate false presumptions, they would need to characterize their model, which relies on sensing and analyzing overt behavior, as one of expression recognition rather than emotion recognition.
27. Much of human action is based on automated involuntary processes. Yet, even in such cases, intentionality (i.e., the capability of the mind to be directed to entities external to it) is a more fruitful premise to explain how people end up taking those actions compared to behavior-related stimulus-response explanations (Brentano, 1887; von Wright, 1971).
28. These descriptions are often referred to as "emotion typologies.".
29. The term "negative emotions" is somewhat misleading, as the presence of *negative* emotions (e.g., sadness, anger) can be experienced as positive or appraised as having a positive function in a person's life situation (Power & Dalgleish, 2016).
30. Later in this section, we discuss the possibility of contradictive feelings toward an appraised event.
31. It is also possible to consider the artefact design the other way around by choosing contexts according to the technical methods being used and framing the goal(s) of the emotion task to be congruent with the (technical) methods. Such a design approach can be rightfully called "narrow" since it is technology oriented (Saariluoma et al., 2016). However, the task situation still remains the foundation for the design questions that a supposed technical artefact must address. In other words, *the task setting is always the primary source which the validity of a supposed technical artefact is evaluated against*.
32. See Mohammad (2022, pp. 243, 248) for a list of emotion tasks that serves as a good starting point for analyzing possible emotion tasks.
33. For further elaboration, see note 28 on technology-oriented approaches.
34. The psychological context is best understood by framing it as "user psychology" (Saariluoma & Oulasvirta, 2010), which helps explain psychological phenomena related to technology use (i.e., people as users or targets of technology use).
35. This claim holds true, though in a more limited manner, to situations in which textual information is used, as people may not recognize what elicited the emotion and may therefore be incapable of expressing it textually (Paavilainen, 2020; Power & Dalgleish, 2016).
36. In this field, there is always a danger of starting to use overt responses as explanations of mental states. Even in the best-case scenario, such responses should be considered as *possible indicators* that are explained by known mechanisms. If this important nuance is lost, epistemic statements can easily become ontological ones.
37. An overlap in evaluated studies between the two articles is possible but not considered here.
38. (1) novelty or unexpectedness, (2) (intrinsic and extrinsic) valence, (3) agency, (4) control or power, and (5) fairness (Scherer, 2009; Scherer & Moors, 2019).
39. It is therefore considered one of the two *core affect* dimensions, of which the other is arousal (Barrett et al., 2019; Russell, 2003). Notably, in the *constructionist* tradition underlying the idea of *core affect*, valence is considered an irreducible variant in people's experiences.
40. Self-congruence refers to congruence or incongruence of an event with one's self-concept.
41. The notion of mixed feelings has mostly been discussed on the conceptual level of emotion themes (Oatley et al., 2006).
42. Researchers have presented hypotheses (and a few experiments) concerning whether similar distinct response effects in facial muscles can be found for the even earlier valenced appraisal criteria of novelty and goal relevancy and for the subsequent appraisal of prospective control (Aue et al., 2007; Scherer & Moors, 2019).
43. Zhang and Arandjelović's work lacks a valid psychological framework, but it analyzes important and interesting engineering issues related to building overt automated emotion inference with the help of image analysis.
44. Such idea often implicitly presumes that technical artefacts, and AI artefacts as their present-day "crown jewel," are *infallible* despite always resulting from human action (Jasanoff, 2016; Roemmich et al., 2023; Stark, 2023). The absurdity of this belief stems from explicitly considering people as the *fallible entity* and point of comparison to prove the infallibility of technology (Roemmich et al., 2023).

45. Sentimentalism usually does not limit the discussion to sentiments attached to explicit propositions but holds that ineffable judgements of events (intuition) are equally if not more purely moral than explicit moral discourse (Haidt, 2001, 2003).
46. As a crude example, the role of safety and intrinsic pleasantness may differ when interpreting a *dangerous event* in a traffic setting versus extreme sports.
47. This *consensus* should not be thought of as a result of social *reflective equilibrium* (Rawls, 1971) but rather a consequence of messy discourse, where power dynamics largely define whose voice is heard and given weight (Jasanoff, 2015, 2016).
48. It is questionable whether sympathy or empathy should be considered as emotion themes or as dispositions to consider the experiences of others.
49. It is also not discussed in enough detail how expressions guide the selection of correcting norm representations that guide the future actions of the agent, thus leaving the selection of norms as a random process of trial and error. Learning norms by trial and error is possible for humans since our action selection is guided by an understanding of nuanced social cues that convey the *relevance* of action and norm representations. However, as AI artefacts rely on formal information processing, they are not capable of a similar type of narrowing down of the *search space* according to relevance (Haaparanta, 2019; Saariluoma & Rauterberg, 2016). This means an endless trial and error of *proper* norm representations, which is not reasonable in real social interactions.
50. The field of computational modeling of emotions is expansive (Battaglino, 2013; Battaglino & Damiano, 2015; Cowen et al., 2019; Ohja et al., 2021; Scherer et al., 2021) and beyond the scope of this article.
51. In the case of Gloor (2022), the *revealed* information is communicated back to the person herself.
52. Contemporary reinforcement learning in automated emotion inference artefacts, where individuals can *correct* a false inference, are not examples of such design because they are based on *forced choice*, which seemingly supports the false presumptions underlying the models and thus does not support proper communication of the individual.
53. Automated emotion inference is always psychologically invasive, as it seeks to recognize hints about a person's internal world without the person actively disclosing them.
54. For an interesting account of such high-stake situations, see (Todorov, 2017).
55. However, as Monteith et al. (2022) well argue, the same bias has different degrees of negative effects to different people.
56. Even examples of reinforced learning that takes the self-reports of individuals into consideration are not flexible in the same manner as human thinking. Since the possibility of correction in the reinforced models relies on the same operationalized presumptions that caused the mistake in the first place, the *corrective self-report* is a case of invalid forced choice (e.g., Gloor, 2022).
57. Currently, in most cases, operationalization involves technical boundaries to which other relevant areas of reasoning must submit, leading to scientifically invalid and ethically questionable applications.

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References

Ajunwa, I. (2020). The paradox of automation as anti-bias intervention. *Cardozo Law Review*, *41*(5), 1671. https://doi.org/10.2139/ssrn.2746078

Ajunwa, I. (2022). Automated video interviewing as the new phrenology. *Berkeley Technology Law Journal*, *36*(3), 101. https://ssrn.com/abstract=3889454

Ambady, N., & Rosenthal, R. (1992). Thin slices of expressive behavior as predictors of interpersonal consequences: A meta-analysis. *Psychological Bulletin*, *111*(2), 256–274. https://doi.org/10.1037/0033-2909.111.2.256

Anderson, M., & Anderson, S. L. (2011). *Machine ethics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511978036

Araño, K. A., Gloor, P., Orsenigo, C., & Vercellis, C. (2021). When old meets new: Emotion recognition from speech signals. *Cognitive Computation*, *13*(3), 771–783. https://doi.org/10.1007/s12559-021-09865-2

Arnold, M. B., & Gasson, J. A. (1954). Feelings and emotions as factors in personality integration. In M. B. Arnold & J. A. Gasson (Eds.), *The human person*. The Ronald Press.

Arnold, M. B. (1969). Human emotion and action. In: T. Mischel (Ed.), *Human action* (pp. 167–197). Academic Press.

Aue, T., Flykt, A., & Scherer, K. R. (2007). First evidence for differential and sequential efferent effects of stimulus relevance and goal conduciveness appraisal. *Biological Psychology*, *74*(3), 347–357. https://doi.org/10.1016/j.biopsycho.2006.09.001

Aue, T., & Scherer, K. R. (2008). Appraisal-driven somatovisceral response patterning: Effects of intrinsic pleasantness and goal conduciveness. *Biological Psychology*, *79*(2), 158–164. https://doi.org/10.1016/j.biopsycho.2008.04.004

Aue, T., & Scherer, K. R. (2011). Effects of intrinsic pleasantness and goal conduciveness appraisals on somatovisceral responding: Somewhat similar, but not identical. *Biological Psychology*, *86*(1)2011), 65–73. https://doi.org/10.1016/j.biopsycho.2010.10.008

Bach, T. A., Khan, A., Hallock, H., Beltrão, G., & Sousa, S. (2024). A systematic literature review of user trust in AI-enabled systems: An HCI perspective. *International Journal of Human–Computer Interaction*, *40*(5), 1251–1266. https://doi.org/10.1080/10447318.2022.2138826

Balkenius, C., Cañamero, L., Pärnamets, P., Johansson, B., Butz, M. V., & Olsson, A. (2016). Outline of a sensorimotor perspective on intrinsically moral agents. *Adaptive Behavior*, *24*(5), 306–319. https://doi.org/10.1177/1059712316667203

Bandura, A. (1991). Social cognitive theory of moral thought and action. In W. M. Kurtines & J. L. Gewirtz (Eds.), *Handbook of moral behavior and development* (Vol. 1, pp. 45–103). Erlbaum.

Bannon, L. (2011). Reimagining HCI: Toward a more human-centered perspective. *Interactions*, *18*(4), 50–57. https://doi.org/10.1145/1978822.1978833

Bardzell, S., & Bardzell, J. (2011). Towards a feminist HCI methodology: Social science, feminism, and HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)* (pp. 675–684). Association for Computing Machinery. https://doi.org/10.1145/1978942.1979041

Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest: a Journal of the American Psychological Society*, *20*(1), 1–68. https://doi.org/10.1177/1529100619832930

Battaglino, C. (2013). Agents with a moral dimension. In *Proceedings of the 2013 International Conference on Autonomous Agents and Multi-agent Systems (AAMAS '13)* (pp. 1421–1422). International Foundation for Autonomous Agents and Multiagent Systems.

Battaglino, C., & Damiano, R. (2015, May 2015). Coping with moral emotions. In R. Bordini, E. Elkind, G. Weiss, & P. Yolum (Eds.), *Proceedings of the 14th International Conference on Autonomous Agents and Multi-agent Systems (AAMAS '15)* (pp. 4–8). International Foundation for Autonomous Agents and Multiagent Systems.

Ben, X., Ren, Y., Zhang, J., Wang, S., Kpalma, K., Meng, W., & Liu, Y. (2022). Video-based facial micro-expression analysis: A survey of datasets, features and algorithms. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(9), 5826–5846. https://doi.org/10.1109/TPAMI.2021.3067464

Benedetti, J. (2005). *Stanislavski: An introduction, revised and updated*. Routledge.

Bernstein, R. J. (1983). *Beyond objectivism and relativism: Science, hermeneutics, and praxis*. Blackwell.

Birhane, A. (2021). Algorithmic injustice: A relational ethics approach. *Patterns (New York, N.Y.)*, *2*(2), 100205. https://doi.org/10.1016/j.patter.2021.100205

Bødker, S. (2015). Third-wave HCI, 10 years later--participation and sharing. *Interactions*, *22*(5), 24–31. https://doi.org/10.1145/2804405

Bødker, S., & Kyng, M. (2018). Participatory design that matters—facing the big issues. *ACM Transactions on Computer-Human Interaction*, *25*(1), Article 4, 1–31. https://doi.org/10.1145/3152421

Boehner, K., DePaula, R., Dourish, P., & Sengers, P. (2007). How emotion is made and measured. *International Journal of Human-Computer Studies*, *65*(4), 275–291. https://doi.org/10.1016/j.ijhcs.2006.11.016

Boehner, K., DePaula, R., Dourish, P., & Sengers, P. (2005). Affect: From information to interaction. In *Proceedings of the 4th decennial conference on Critical computing: Between sense and sensibility (CC '05)* (pp. 59–68). Association for Computing Machinery. https://doi.org/10.1145/1094562.1094570

Bonnefon, J., Hopfensitz, A., & De Neys, W. (2015). Face-ism and kernels of truth in facial inferences. *Trends in Cognitive Sciences*, *19*(8), 421–422. https://doi.org/10.1016/j.tics.2015.05.002

Bondioli, M., Chessa, S., Kocian, A., & Pelagatti, S. (2024). A survey on technological tools and systems for diagnosis and therapy of autism spectrum disorder. *Human–Computer Interaction*, *39*(3–4), 145–173. https://doi.org/10.1080/07370024.2023.2242345

Boutin, A., Desmoulin-Canselier, S., & Lévêque, L. (2023). On legal and ethical challenges of automatic facial expression recognition: An exploratory study. In *Proceedings of the 2023 ACM International Conference on Interactive Media Experiences (IMX 2023)* (pp. 226–229). ACM. https://doi.org/10.1145/3573381.3597217

Boyd, K. L., & Andalibi, N. (2023). Automated emotion recognition in the workplace: How proposed technologies reveal potential futures of work. In *Proceedings of the ACM on Human-Computer Interaction 7 (1 CSCW)*, Article 95. ACM. https://doi.org/10.1145/3579528

Branford, J. (2024). An (e)affective bind: Situated affectivity and the prospect of affect recognition. *IEEE Transactions on Affective Computing*, *15*(1), 18–29. https://doi.org/10.1109/TAFFC.2023.3281069

Brentano, F. (1887/1995). *Descriptive psychology* (B. Müller, Trans. and Ed.). Routledge.

Brüne, M., & Brüne-Cohrs, U. (2006). Theory of mind—evolution, ontogeny, brain mechanisms and psychopathology. *Neuroscience and Biobehavioral Reviews*, *30*(4), 437–455. https://doi.org/10.1016/j.neubiorev.2005.08.001

Chen, F., Shao, J., Zhu, S., & Shen, H. T. (2023). *Multivariate, Multi-Frequency and Multimodal: Rethinking Graph Neural Networks for Emotion Recognition in Conversation* [Paper presentation]. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, pp. 10761–10770. https://doi.org/10.1109/CVPR52729.2023.01036

Chen, H., Shi, H., Liu, X., Li, X., & Zhao, G. (2023). SMG: A micro-gesture dataset towards spontaneous body gestures for emotional stress state analysis. *International Journal of Computer Vision*, *131*(6), 1346–1366. https://doi.org/10.1007/s11263-023-01761-6

Chen, Z., & Whitney, D. (2019). Tracking the affective state of unseen persons. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(15), 7559–7564. https://doi.org/10.1073/pnas.1812250116

Churchland, P. S., Ramachandran, V. S., & Sejnowski, T. J. (1994). A critique of pure vision. In C. Koch & J. L. Davis (Eds.), *Large-scale neuronal theories of the brain* (pp. 23–60). The MIT Press.

Cockton, G. (2004). Value-centred HCI. In: *Proceedings of the third Nordic conference on Human-computer interaction (NordiCHI '04)* (pp. 149–160). Association for Computing Machinery. https://doi.org/10.1145/1028014.1028038

Coeckelbergh, M. (2010). Moral appearances: Emotions, robots, and human morality. *Ethics and Information Technology*, *12*(3), 235–241. https://doi.org/10.1007/s10676-010-9221-y

Cohen, J. R., Pant, L. W., & Sharp, D. J. (2001). An examination of differences in ethical decision-making between Canadian business students and accounting professionals. *Journal of Business Ethics*, *30*(4), 319–336. https://doi.org/10.1023/A:1010745425675

Corvite, S., Roemmich, K., Rosenberg, T. I., & Andalibi, N. (2023). Data subjects' perspectives on emotion artificial intelligence use in the workplace: A relational ethics lens. In *Proceedings of the ACM on Human-Computer Interaction 7 (1 CSCW)*, Article 124. ACM. https://doi.org/10.1145/3579600

Cowen, A. S., & Keltner, D. (2017). Self-report captures 27 distinct categories of emotion bridged by continuous gradients. *Proceedings of the National Academy of Sciences of the United States of America*, *114*(38), e7900–e7909. https://doi.org/10.1073/pnas.1702247114

Cowen, A., Sauter, D., Tracy, J. L., & Keltner, D. (2019). Mapping the passions: Toward a high-dimensional taxonomy of emotional experience and expression. *Psychological Science in the Public Interest: a Journal of the American Psychological Society*, *20*(1)2019), 69–90. https://doi.org/10.1177/1529100619850176

Crnkovic, G. D., & Çürüklü, B. (2012). Robots: Ethical by design. *Ethics and Information Technology*, *14*(1), 61–71. https://doi.org/10.1007/s10676-011-9278-2

Damasio, A. R. (2006). *Descartes' error: Emotion, reason and the human brain*. Vintage.

Dan Glauser, E. S., & Scherer, K. R. (2008). Neuronal processes involved in subjective feeling emergence: Oscillatory activity during an emotional monitoring task. *Brain Topography*, *20*(4), 224–231. https://doi.org/10.1007/s10548-008-0048-3

Deguchi, A., Hirai, C., Matsuoka, H., Nakano, T., Oshima, K., Tai, M., & Tani, S. (2020). What is society 5.0? In A. Deguchi & O. Kamimura (Eds.), *Society 5.0*. Springer. https://doi.org/10.1007/978-981-15-2989-4_1

Dennett, D. (1988). Quining qualia. In A. Marcel & E. Bisiach (Eds.), *Consciousness in modern science* (pp. 42–77). Oxford University Press.

De Prisco, M., Oliva, V., Fico, G., Montejo, L., Possidente, C., Bracco, L., Fortea, L., Anmella, G., Hidalgo-Mazzei, D., Fornaro, M., de Bartolomeis, A., Serretti, A., Murru, A., Vieta, E., & Radua, J. (2023). Differences in facial emotion recognition between bipolar disorder and other clinical populations: A systematic review and meta-analysis. *Progress in Neuro-Psychopharmacology & Biological Psychiatry*, *127*(2023), 110847. https://doi.org/10.1016/j.pnpbp.2023.110847

de Sousa, R. (2022). Who needs values when we have valuing? Comments on Jean Moritz Müller, the world-directedness of emotional feeling. *Emotion Review: journal of the International Society for Research on Emotion*, *14*(4), 257–261. https://doi.org/10.1177/17540739221085575

Duarte, E. F., & Baranauskas, M. C. (2016). Revisiting the three HCI waves: A preliminary discussion on philosophy of science and research paradigms. In *Proceedings of the 15th Brazilian Symposium on Human Factors in Computing Systems (IHC '16)* (Vol. 38, pp. 1–4). Association for Computing Machinery. https://doi.org/10.1145/3033701.3033740

Eitel-Porter, R. (2021). Beyond the promise: Implementing ethical AI. *AI and Ethics*, *1*(1), 73–80. https://doi.org/10.1007/s43681-020-00011-6

Ekman, P. (1993). Facial expression and emotion. *The American Psychologist*, *48*(4), 384–392. https://doi.org/10.1037/0003-066X.48.4.384
Ekman, P. (1992). An argument for basic emotions. *Cognition and Emotion*, *6*(3–4), 169–200. https://doi.org/10.1080/02699939208411068
Ekman, P., & Friesen, W. V. (1978). *Facial Action Coding System (FACS)*. APA PsycTests. https://doi.org/10.1037/t27734-000
Ekman, P., & Friesen, W. V. (1971). Constants across cultures in the face and emotion. *Journal of Personality and Social Psychology*, *17*(2), 124–129. https://doi.org/10.1037/h0030377
Ellsworth, P. C., & Scherer, K. R. (2003). Appraisal processes in emotion. In R. J. Davidson, K. R. Scherer, & H. H. Goldsmith (Eds.), *Handbook of affective sciences* (pp. 572–595). Oxford University Press.
Fiesler, C., Morrison, S., & Bruckman, A. S. (2016). An archive of their own: A case study of feminist HCI and values in design. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)* (pp. 2574–2585). Association for Computing Machinery. https://doi.org/10.1145/2858036.2858409
Fish, B., & Stark, L. (2021, July). Reflexive design for fairness and other human values in formal models. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 89–99). ACM. https://doi.org/10.1145/3461702.3462518
Folkman, S., & Lazarus, R. S. (1988). Coping as a mediator of emotion. *Journal of Personality and Social Psychology*, *54*(3), 466–475. https://doi.org/10.1037/0022-3514.54.3.466
Folz, J., Fiacchino, D., Nikolić, M., van Steenbergen, H., & Kret, M. E. (2022). Reading your emotions in my physiology? Reliable emotion interpretations in absence of a robust physiological resonance. *Affective Science*, *3*(2), 480–497. https://doi.org/10.1007/s42761-021-00083-5
Frauenberger, C., Rauhala, M., & Fitzpatrick, G. (2017). In-action ethics. *Interacting with Computers*, *29*(2), 220–236. https://doi.org/10.1093/iwc/iww024
Friedman, B., & Hendry, D. (2019). *Value sensitive design: Shaping technology with moral imagination*. The MIT Press. https://doi.org/10.7551/mitpress/7585.001.0001
Frijda, N. H. (1986). *The emotions*. Cambridge University Press.
Frijda, N. H., Kuipers, P., & Schure, E. (1989). Relations among emotion, appraisal, and emotional action readiness. *Journal of Personality and Social Psychology*, *57*(2), 212–228. https://doi.org/10.1037/0022-3514.57.2.212
Fu, H., & Lyu, Y. (2025). Implementing Facial Recognition Technology in a University Setting: An Organizational Justice Perspective. *International Journal of Human–Computer Interaction*, *41*(15), 9248–9261. https://doi.org/10.1080/10447318.2024.2423352
Fuchsberger, V., & Frauenberger, C. (2025). Doing responsibilities in entangled worlds. *Human–Computer Interaction*, *40*(1–4), 65–88. https://doi.org/10.1080/07370024.2023.2269934
Gaver, W. (2009). Designing for emotion (among other things). *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *364*(1535), 3597–3604. https://doi.org/10.1098/rstb.2009.0153
Gentsch, K., Beermann, U., Wu, L., Trznadel, S., & Scherer, K. R. (2020). Temporal unfolding of micro-valences in facial expression evoked by visual, auditory, and olfactory stimuli. *Affective Science*, *1*(4), 208–224. https://doi.org/10.1007/s42761-020-00020-y
Gentsch, K., Grandjean, D., & Scherer, K. R. (2015). Appraisals generate specific configurations of facial muscle movements in a gambling task: Evidence for the component process model of emotion. *PloS One*, *10*(8), e0135837. https://doi.org/10.1371/journal.pone.0135837
Giaccardi, E., Redström, J., & Nicenboim, I. (2025). The making(s) of more-than-human design: Introduction to the special issue on more-than-human design and HCI. *Human–Computer Interaction*, *40*(1–4), 1–16. https://doi.org/10.1080/07370024.2024.2353357
Giddens, A. (1984). *The constitution of society: Outline of the theory of structuration*. Polity Press.
Giner-Sorolla, R. (1999). Affect in attitude: Immediate and deliberative perspectives. In S. Chaiken and Y. Trope (Eds.), *Dual-process theories in social psychology* (pp. 441–461). The Guilford Press.
Gillespie, T. (2019). *Systems engineering for ethical autonomous systems*. SciTech Publishing.
Gitelman, L. (Ed.) (2013). *"Raw Data" is an oxymoron*. The MIT Press.
Gloor, P. (2022). *Happimetrics - leveraging AI to untangle the surprising link between ethics, happiness and business success*. Edward Elgar Publishing.
Goffman, E. (1979). *Gender advertisements*. Harper & Row Publishers.
Goffman, E. (1956). *The presentation of self in everyday life.* University of Edinburgh, Social Sciences Research Centre.
Gomila, A., & Amengual, A. (2009). Moral emotions for autonomous agents. In J. Vallverdú (Ed.), *Handbook of research on synthetic emotions and sociable robotics: New applications in affective computing and artificial intelligence* (pp. 166–179). Information Science Reference. https://doi.org/10.4018/978-1-60566-354-8.ch010
Gorman, E. (2005). Gender stereotypes, same-gender preferences, and organizational variation in the hiring of women: Evidence from law firms. *American Sociological Review*, *70*(4), 702–728. https://doi.org/10.1177/000312240507000408
Graff, J. (2024). Moral sensitivity and the limits of artificial moral agents. *Ethics and Information Technology*, *26*(1), 13. https://doi.org/10.1007/s10676-024-09755-9
Graham, G. (2023). Behaviorism. In Edward N. Zalta & Uri Nodelman (Eds.), *The Stanford Encyclopedia of Philosophy (*Spring 2023 Edition*)*. https://plato.stanford.edu/archives/spr2023/entries/behaviorism/

Gratch, J., Greene, G., Picard, R., Urquhart, L., & Valstar, M. (2024). Guest editorial: Ethics in affective computing. *IEEE Transactions on Affective Computing*, *15*(1), 1–3. https://doi.org/10.1109/TAFFC.2023.3322918

Gross, J. J. (2008). Emotion and emotion regulation: Personality processes and individual differences. In O. P. John, R. W. Robins, & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (3rd ed., pp. 701–724). The Guilford Press.

Guerrero, G., Avila, D., da Silva, F. J. M., Pereira, A., & Fernández-Caballero, A. (2023). Internet-based identification of anxiety in university students using text and facial emotion analysis. *Internet Interventions*, *34*, 100679. https://doi.org/10.1016/j.invent.2023.100679

Gunkel, D. J. (2018). The relational turn: Third wave HCI and phenomenology. In M. Filimowicz & V. Tzankova (Eds.), *New directions in third wave human-computer interaction: Volume 1 - technologies. Human–computer interaction series*. Springer. https://doi.org/10.1007/978-3-319-73356-2_2

Guo, W., & Wang, Y. (2024). Convolutional gated recurrent unit-driven multidimensional dynamic graph neural network for subject-independent emotion recognition. *Expert Systems with Applications*, *238*(Part B), 121889. https://doi.org/10.1016/j.eswa.2023.121889

Görnemann, E., & Spiekermann, S. (2024). Emotional Responses to Human Values in Technology: The Case of Conversational Agents. *Human–Computer Interaction*, *39*(5-6), 310–337. vember. https://doi.org/10.1080/07370024.2022.2136094

Haaparanta, L. (2019). *Rajan taju: Filosofisia esseitä* [Understanding demarcation lines: Philosophical essays]. Gaudeamus.

Habermas, J. (1995). *Moral consciousness and communicative action*. (Repr.). Polity.

Habermas, J. (1981). *Theorie des kommunikativen Handelns: Band 1, Handlungsrationalität und gesellschaftliche Rationalisierung [The theory of communicative action: Vol. 1, Reason and the rationalization of society]*. Suhrkamp.

Haidt, J. (2008). Morality. *Perspectives on Psychological Science: a Journal of the Association for Psychological Science*, *3*(1), 65–72. https://doi.org/10.1111/j.1745-6916.2008.00063.x

Haidt, J. (2003). The moral emotions. In R. J. Davison, K. R. Scherer, & H. H. Goldsmith (Eds.), *Handbook of affective sciences* (pp. 852–870). Oxford University Press. https://www.overcominghateportal.org/uploads/5/4/1/5/5415260/the_moral_emotions.pdf

Haidt, J. (2001). The emotional dog and its rational tail: A social intuitionist approach to moral judgment. *Psychological Review*, *108*(4), 814–834. https://doi.org/10.1037/0033-295X.108.4.814

Harper, R. H. R. (2019). The role of HCI in the age of AI. *International Journal of Human–Computer Interaction*, *35*(15), 1331–1344. https://doi.org/10.1080/10447318.2019.1631527

Harrison, S., Sengers, P., & Tatar, D. (2011). Making epistemological trouble: Third-paradigm HCI as successor science. *Interacting with Computers*, *23*(5), 385–392. https://doi.org/10.1016/j.intcom.2011.03.005

Harrison, S., Tatar, D., & Sengers, P. (2007). The three paradigms of HCI. In *Alt. Chi. Session at the SIGCHI Conference on human factors in computing systems (CHI '07), San Jose, California (*pp. 1–18). Association for Computing Machinery.

Heaven, D. (2020). Why faces don't always tell the truth about feelings. *Nature*, *578*(7796), 502–504. https://doi.org/10.1038/d41586-020-00507-5

Hindriks, F., & Sauer, H. (2020). The mark of the moral: Beyond the sentimentalist turn. *Philosophical Psychology*, *33*(4), 569–591. https://doi.org/10.1080/09515089.2020.1731444

Hoegen, J., Lucas, G., Shore, D., Parkinson, B., & Gratch, J. (2023). *How expression and context determine second-person judgments of emotion* [Paper presentation]. 11th International Conference on Affective Computing and Intelligent Interaction (ACII), Cambridge, MA, 1–7. https://doi.org/10.1109/ACII59096.2023.10388189

Hofstede, G. (1980). *Culture's consequences*. Sage Publications.

Holst, J. (2021). Ethical rationality in AI: On the prospect of becoming a full ethical agent. In S. Thompson (Ed.), *Machine law, ethics, and morality in the age of artificial intelligence* (pp. 69–84). IGI Global. https://doi.org/10.4018/978-1-7998-4894-3.ch005

Holtzman, G. S. (2018). A neuropsychological challenge to the sentimentalism/rationalism distinction. *Synthese*, *195*(5), 1873–1889. https://doi.org/10.1007/s11229-017-1344-9

Hornbæk, K. (2006). Current practice in measuring usability: Challenges to usability studies and research. *International Journal of Human-Computer Studies*, *64*(2), 79–102. https://doi.org/10.1016/j.ijhcs.2005.06.002

Howell, N., Devendorf, L., Alfonso, T., Gálvez, V., Tian, R., & Ryokai, K. (2018). Tensions of data-driven reflection: A case study of real-time emotional biosensing. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)* (Paper 431, pp. 1–13). Association for Computing Machinery. https://doi.org/10.1145/3173574.3174005

Hult, L., Irestig, M., & Lundberg, J. (2006). Design perspectives. *Human–Computer Interaction*, *21*(1), 5–48. https://doi.org/10.1207/s15327051hci2101_2

Hume, D. (1972). *A treatise of human nature*. Dent.

Höök, K. (2008). Affective loop experiences – what are they? In H. Oinas-Kukkonen, P. Hasle, M. Harjumaa, K. Segerståhl, & P. Øhrstrøm (Eds.), *Persuasive Technology. PERSUASIVE 2008. Lecture Notes in Computer Science* (Vol. 5033). Springer. https://doi.org/10.1007/978-3-540-68504-3_1

Höök, K., Ståhl, A., Sundström, P., & Laaksolaahti, J. (2008). Interactional empowerment. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)* (pp. 647–656). Association for Computing Machinery. https://doi.org/10.1145/1357054.1357157

IEEE (2019). *White paper - ethically aligned design - a vision for prioritizing human well-being with autonomous and intelligent systems*. IEEE.

Jasanoff, S. (2016). *The ethics of invention—technology and the human future*. W. W. Norton & Company, Inc.

Jasanoff, S. (2015). Future imperfect: Science, technology and the imaginations of modernity. In S. Jasanoff & S. Kim (Eds.), *Dreamscapes of modernity: Sociotechnical imaginaries and the fabrication of power* (pp. 1–33). The University of Chicago Press.

Jeon, M. (2017). Emotions and affect in human factors and human–computer interaction: Taxonomy, theories, approaches, and methods. In M. Jeon (Ed.), *Emotions and affect in human factors and human-computer interaction* (pp. 3–26). Academic Press. ISBN 9780128018514. https://doi.org/10.1016/B978-0-12-801851-4.00001-X

Jokinen, J. P. P., & Silvennoinen, J. (2020). The appraisal theory of emotion in human–computer interaction. In R. Rousi, J. Leikas, and P. Saariluoma (Eds.), *Emotions in technology design: From experience to ethics. Human - computer interaction series* (pp. 27–39). Springer International Publishing. https://doi.org/10.1007/978-3-030-53483-7_3

Jones, D., & Gregor, S. (2007). The anatomy of a design theory. *Journal of the Association for Information Systems*, *8*(5), 312–335. https://doi.org/10.17705/1jais.00129

Kant, I. (1781/1976). *Kritik der reinen Vernunft* [The critique of pure reason]. Felix Meiner.

Kaplan, A. (1964). *The conduct of inquiry: Methodology for behavioral science*. Chandler Publishing Company; Harper & Row.

Katirai, A. (2024). Ethical considerations in emotion recognition technologies: A review of the literature. *AI and Ethics*, *4*(4), 927–948. https://doi.org/10.1007/s43681-023-00307-3

Kauppinen, A. (2022). Moral sentimentalism. In E. Zalta (Ed.), *The Stanford encyclopedia of philosophy*. (Spring 2022 Edition). https://plato.stanford.edu/archives/spr2022/entries/moral-sentimentalism/

Keltner, D., Brooks, J. A., & Cowen, A. (2023). Semantic space theory: Data-driven insights into basic emotions. *Current Directions in Psychological Science*, *32*(3), 242–249. https://doi.org/10.1177/09637214221150511

Ketokivi, M. (2015). *Tilastollinen päättely ja tieteellinen argumentointi [Statistical inference and scientific argumentation]* (2nd ed.). Gaudeamus.

Khare, S. K., Blanes-Vidal, V., Nadimi, E. S., & Acharya, U. R. (2024). Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations. *Information Fusion*, *102*, 102019. https://doi.org/10.1016/j.inffus.2023.102019

Koshiyama, A., Kazim, E., Treleaven, P., Rai, P., Szpruch, L., Pavey, G., Ahamat, G., Leutner, F., Goebel, R., Knight, A., Adams, J., Hitrova, C., Barnett, J., Nachev, P., Barber, D., Chamorro-Premuzic, T., Klemmer, K., Gregorovic, M., Khan, S., … Chatterjee, S. (2024). Towards algorithm auditing: Managing legal, ethical and technological risks of AI, ML and associated algorithms. *Royal Society Open Science*, *11*(5), 230859. https://doi.org/10.1098/rsos.230859

Kosti, R., Alvarez, J. M., Recasens, A., & Lapedriza, A. (2020). Context based emotion recognition using EMOTIC dataset. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *42*(11), 2755–2766. https://doi.org/10.1109/TPAMI.2019.2916866

Kreibig, S. D. (2010). Autonomic nervous system activity in emotion: A review. *Biological Psychology*, *84*(3), 394–421. https://doi.org/10.1016/j.biopsycho.2010.03.010

Kuhn, T. S. (1996). *The structure of scientific revolutions* (3rd ed.). University of Chicago Press.

Kuppens, P., Van Mechelen, I., & Rijmen, F. (2008). Toward disentangling sources of individual differences in appraisal and anger. *Journal of Personality*, *76*(4), 969–1000. https://doi.org/10.1111/j.1467-6494.2008.00511.x

Kuppens, P., Van Mechelen, I., Smits, D. J. M., De Boeck, P., & Ceuleman, E. (2007). Individual differences in patterns of appraisal and anger experience. *Cognition & Emotion*, *21*(4), 689–713. https://doi.org/10.1080/02699930600859219

Kuutti, K., & Bannon, L. J. (2014). The turn to practice in HCI: Towards a research agenda. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)* (pp. 3543–3552). Association for Computing Machinery. https://doi.org/10.1145/2556288.2557111

Laakasuo, M., Sundvall, J. R., Berg, A., Drosinou, M., Herzon, V., Kunnari, A., Koverola, M., Repo, M., Saikkonen, T., & Palomäki, J. (2021). Moral psychology and artificial agents (part two): The transhuman connection. In S. Thompson (Ed.), *Machine law, ethics, and morality in the age of artificial intelligence* (pp. 189–204). IGI Global. https://doi.org/10.4018/978-1-7998-4894-3.ch011

Lee, C., Chaspari, T., Provost, E. M., & Narayanan, S. S. (2023). An engineering view on emotions and speech: From analysis and predictive models to responsible human-centered applications. *Proceedings of the IEEE*, *111*(10), 1–17. https://doi.org/10.1109/JPROC.2023.3276209

Leibniz, G. (1704/1981). *New essays on human understanding*. Cambridge University Press.

Leslie, A. M., Friedman, O., & German, T. P. (2004). Core mechanisms in 'theory of mind. *Trends in Cognitive Sciences*, *8*(12), 528–533. https://doi.org/10.1016/j.tics.2004.10.001

Liu, Y., Moore, A., Webb, J., & Vallor, S. (2022). Artificial moral advisors: A new perspective from moral psychology. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (AIES '22)* (pp. 436–445). Association for Computing Machinery. https://doi.org/10.1145/3514094.3534139

Liu, J (2021). Why we need emotional intelligence in the design of autonomous social robots and how Confucian moral sentimentalism can help. In H. Li., et al. (Eds.), *Social Robotics (ICSR 2021), lecture notes in computer science* (Vol. 13086). Springer. https://doi.org/10.1007/978-3-030-90525-5_20

Maldonato, M., & Valerio, P. (2018). Artificial entities or moral agents? How AI is changing human evolution. In A. Esposito, M. Faudez-Zanuy, F. Morabito, & E. Pasero (Eds.), *Multidisciplinary approaches to neural computing. Smart innovation, systems and technologies* (Vol. 69). Springer. https://doi.org/10.1007/978-3-319-56904-8_36

Malle, B. F. (2021). Moral judgments. *Annual Review of Psychology*, *72*(1), 293–318. https://doi.org/10.1146/annurev-psych-072220-104358

Malle, B. F. (2011). Attribution theories: How people make sense of behavior. In D. Chadee (Ed.), *Theories in social psychology* (pp. 72–95). Wiley-Blackwell.

Malle, B. F., Austerweil, J. L., Chi, V. B., Kenett, Y., Beck, E. D., Thapa, S., & Allaham, M. (2021). Cognitive properties of norm representations. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. https://escholarship.org/uc/item/0cr0x2b3

Malle, B. F., Rosen, E., Chi, V. B., Berg, M., & Haas, P. (2020). *A general methodology for teaching norms to social robots* [Paper presentation]. In The 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2020, Naples, Italy, 1395–1402. https://doi.org/10.1109/RO-MAN47096.2020.9223610

Malle, B. F., & Scheutz, M. (2019). Learning how to behave. In O. Bendel (Ed.), *Handbuch Maschinenethik*. Springer VS. https://doi.org/10.1007/978-3-658-17483-5_17

Markets and Markets (2023). Emotion detection and recognition market size, share and global market forecast to 2027. *MarketsandMarkets*. https://www.marketsandmarkets.com/Market-Reports/emotion-detection-recognition-market-23376176.html

McAuliffe, W. H. B. (2019). Do emotions play an essential role in moral judgments? *Thinking & Reasoning*, *25*(2), 207–230. https://doi.org/10.1080/13546783.2018.1499552

McStay, A. (2020). Emotional AI, soft biometrics and the surveillance of emotional life: An unusual consensus on privacy. *Big Data & Society*, *7*(1), 205395172090438. https://doi.org/10.1177/2053951720904386

McStay, A. (2020). Emotional AI and EdTech: Serving the public good? *Learning, Media and Technology*, *45*(3), 270–283. https://doi.org/10.1080/17439884.2020.1686016

McStay, A., & Rosner, G. (2021). Emotional artificial intelligence in children's toys and devices: Ethics, governance and practical remedies. *Big Data & Society*, *8*(1), 205395172199487. https://doi.org/10.1177/2053951721994877

Meinhardt, J., Sodian, J., Thoermer, C., Döhnel, K., & Sommer, M. (2011). True- and false-belief reasoning in children and adults: An event-related potential study of theory of mind. *Developmental Cognitive Neuroscience*, *1*(1), 67–76. ISSN 1878-9293. https://doi.org/10.1016/j.dcn.2010.08.001

Menary, R. (2010). Introduction to the special issue on 4E cognition. *Phenomenology and the Cognitive Sciences*, *9*(4), 459–463. https://doi.org/10.1007/s11097-010-9187-6

Mertala, P., Moens, E., & Teräs, M. (2022). Highly cited educational technology journal articles: A descriptive and critical analysis. *Learning, Media and Technology*, *49*(2), 216–229. https://doi.org/10.1080/17439884.2022.2141253

Mischel, W., & Shoda, Y. (1995). A cognitive–affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. *Psychological Review*, *102*(2), 246–268. https://doi.org/10.1037/0033-295X.102.2.246

Michell, J. (1997). Quantitative science and the definition of measurement in psychology. *British Journal of Psychology*, *88*(3), 355–383. https://doi.org/10.1111/j.2044-8295.1997.tb02641.x

Miller, G. (2003). The cognitive revolution: A historical perspective. *Trends in Cognitive Sciences*, *7*(3), 141–144. https://doi.org/10.1016/s1364-6613(03)00029-9

Mobbs, D., Weiskopf, N., Lau, H. C., Featherstone, E., Dolan, R. J., & Frith, C. D. (2006). The Kuleshov effect: The influence of contextual framing on emotional attributions. *Social Cognitive and Affective Neuroscience*, *1*(2), 95–106. https://doi.org/10.1093/scan/nsl014

Mohammad, S. M. (2022). Ethics sheet for automatic emotion recognition and sentiment analysis. *Computational Linguistics*, *48*(2), 239–278. https://doi.org/10.1162/coli_a_00433

Monroe, A. E., & Malle, B. F. (2019). People systematically update moral judgments of blame. *Journal of Personality and Social Psychology*, *116*(2), 215–236. https://doi.org/10.1037/pspa0000137

Monteith, S., Glenn, T., Geddes, J., Whybrow, P. C., & Bauer, M. (2022). Commercial use of emotion artificial intelligence (AI): Implications for psychiatry. *Current Psychiatry Reports*, *24*(3), 203–211. https://doi.org/10.1007/s11920-022-01330-7

Moors, A., Ellsworth, P. C., Scherer, K. R., & Frijda, N. H. (2013). Appraisal theories of emotion: State of the art and future development. *Emotion Review*, *5*(2), 119–124. https://doi.org/10.1177/1754073912468165

Mumenthaler, C., & Sander, D. (2015). Automatic integration of social information in emotion recognition. *Journal of Experimental Psychology. General*, *144*(2), 392–399. https://doi.org/10.1037/xge0000059

Müller, V. C. (2020). Ethics of artificial intelligence and robotics. In E. N. Zalta (Ed.), *Stanford encyclopedia of philosophy* (pp. 1–70). CSLI Stanford University.

Myllylä, M. T., & Saariluoma, P. (2022). Expertise and becoming conscious of something. *New Ideas in Psychology*, *64*, 100916. https://doi.org/10.1016/j.newideapsych.2021.100916

Nakamura, K. (2019). My algorithms have determined you're not human: AI-ML, reverse Turing-tests, and the disability experience. In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19), 2019, Pittsburgh, PA, USA* (pp. 1–2). Association for Computing Machinery. https://doi.org/10.1145/3308561.3353812

Nallur, V., & Finlay, G. (2023). Empathetic AI for ethics-in-the-small. *AI & Society*, *38*(2), 973–974. https://doi.org/10.1007/s00146-022-01466-3

Neal, T., Slobogin, C., Saks, M. J., Faigman, D. L., & Geisinger, K. F. (2019). Psychological assessments in legal contexts: Are courts keeping "junk science" out of the courtroom? *Psychological Science in the Public Interest: a Journal of the American Psychological Society*, *20*(3), 135–164. https://doi.org/10.1177/1529100619888860

Norman, D. A. (1968). Toward a theory of memory and attention. *Psychological Review*, *75*(6), 522–536. https://doi.org/10.1037/h0026699

Nussbaum, M. C. (2004). *Hiding from humanity: Disgust, shame, and the law*. Princeton University Press.

Oatley, K., Keltner, D., & Jenkins, J. (2006). *Understanding emotions*. Blackwell Publishing.

Ojha, S., Vitale, J., & Williams, M. A. (2021). Computational emotion models: A thematic review. *International Journal of Social Robotics*, *13*(6), 1253–1279. https://doi.org/10.1007/s12369-020-00713-1

Ozmen Garibay, O., Winslow, B., Andolina, S., Antona, M., Bodenschatz, A., Coursaris, C., Falco, G., Fiore, S. M., Garibay, I., Grieman, K., Havens, J. C., Jirotka, M., Kacorri, H., Karwowski, W., Kider, J., Konstan, J., Koon, S., Lopez-Gonzalez, M., Maifeld-Carucci, I., … Xu, W. (2023). Six human-centered artificial intelligence grand challenges. *International Journal of Human–Computer Interaction*, *39*(3), 391–437. https://doi.org/10.1080/10447318.2022.2153320

Paavilainen, P. (2020). *Kuinka ajatukset syntyvät: Aivotutkimuksen ajankohtaisia kysymyksiä* [How thoughts are born: Current issues in brain research]. Gaudeamus.

Picard, R. W. (1997). *Affective computing*. MIT Press.

Pigden, C. R. (Ed.). (2010). *Hume on is and ought*. Macmillan.

Pink, S., Leder Mackley, K. L., Mitchell, V., Hanratty, M., Escobar-Tello, C., Bhamra, T., & Morosanu, R. (2013). Applying the lens of sensory ethnography to sustainable HCI. *ACM Transactions on Computer-Human Interaction*, *20*(4), 1–18. https://doi.org/10.1145/2494261

Plutchik, R. (1980). Chapter 1 - A general psychoevolutionary theory of emotion. In H. Kellerman and R. Plutchik (Eds.), *Emotion: Theory, Research and Experience*., Vol. 1, Theories of Emotion (pp. 3–33). Academic Press. https://doi.org/10.1016/B978-0-12-558701-3.50007-7

Power, M. J., & Dalgleish, T. (2016). *Cognition and emotion: From order to disorder* (3rd. ed.). Psychology Press.

Prabowo, D. W., Nugroho, H. A., Setiawan, N. A., & Debayle, J. (2023). A systematic literature review of emotion recognition using EEG signals. *Cognitive Systems Research*, *82*, 101152. https://doi.org/10.1016/j.cogsys.2023.101152

Rawls, J. (1971). *A theory of justice*. Belknap Press of Harvard University Press.

Robert, L. P., Pierce, C., Marquis, L., Kim, S., & Alahmad, R. (2020). Designing fair AI for managing employees in organizations: A review, critique, and design agenda. *Human–Computer Interaction*, *35*(5–6), 545–575. https://doi.org/10.1080/07370024.2020.1735391

Roemmich, K., Rosenberg, T., Fan, S., & Andalibi, N. (2023). Values in emotion artificial intelligence hiring services: Technosolutions to organizational problems. *Proceedings of the ACM on Human-Computer Interaction*, *7*(1 CSCW), Article 109, 28. https://doi.org/10.1145/3579543

Roseman, I. J. (1984). Cognitive determinants of emotion: A structural theory. In P. Shaver (Ed.), *Review of personality and social psychology*, (Vol. 5, pp. 11–36). Sage.

Ross, L., & Nisbett, R. E. (1991). *The person and the situation: Perspectives of social psychology*. McGraw–Hill Book Company.

Rousi, R., Leikas, J., & Saariluoma, P. (2020). *Emotions in technology design: Emotions to ethics*. Springer International Publishing AG.

Rousi, R. (2013). The experience of no experience Elevator UX and the role of unconscious experience. In *Proceedings of International Conference on Making Sense of Converging Media (AcademicMindTrek '13)* (pp. 289–292). Association for Computing Machinery. https://doi.org/10.1145/2523429.2523455

Ruckenstein, M., Velkova, J. (2019). Automating Society Report (2019) – Finland. *Algorithm Watch*. https://algorithmwatch.org/en/automating-society-2019/finland

Russell, J. A. (2003). Core affect and the psychological construction of emotion. *Psychological Review*, *110*(1), 145–172. https://doi.org/10.1037/0033-295X.110.1.145

Russell, J. A., & Barrett, L. F. (1999). Core affect, prototypical emotional episodes, and other things called emotion: Dissecting the elephant. *Journal of Personality and Social Psychology*, *76*(5), 805–819. https://doi.org/10.1037/0022-3514.76.5.805

Russell, P. S., & Giner-Sorolla, R. (2011). Social justifications for moral emotions: When reasons for disgust are less elaborated than for anger. *Emotion (Washington, D.C.)*, *11*(3), 637–646. https://doi.org/10.1037/a0022600

Russell, S., & Norvig, P. (2016). *Artificial intelligence: A modern approach* (3rd ed.). Pearson Education Limited.

Rydenfelt, H. (2015). Emotional interpretants and ethical inquiry. *Sign Systems Studies*, *43*(4), 501–517. https://doi.org/10.12697/SSS.2015.43.4.08

Saariluoma, P. (2021). Hume's guillotine and intelligent technologies. *Human-Intelligent Systems Integration*, *3*(3), 241–250. https://doi.org/10.1007/s42454-021-00035-1
Saariluoma, P. (2005). Explanatory frameworks for interaction design. In A. Pirhonen, P. Saariluoma, H. Isomäki, & C. Roast (Eds.), *Future interaction design*. Springer. https://doi.org/10.1007/1-84628-089-3_5
Saariluoma, P. (2003). *Ajattelu työelämässä: Erehdyksistä mahdollisuuksiin* [Thinking in worklife: From mistakes to opportunities]. WSOY.
Saariluoma, P. (1997). *Foundational analysis: Presuppositions in experimental psychology*. Routledge.
Saariluoma, P. (1995). *Chess players' thinking: A cognitive psychological approach*. Routledge.
Saariluoma, P. (1990). Apperception and restructuring in chess players' problem solving. In K. J. Gilhooly, M. T G. Keane, R. H. Logie, & G. Erdos (Eds.), *Lines of thought: Reflections on the psychology of thinking* (pp. 41–57). Wiley.
Saariluoma, P., Cañas, J., & Leikas, J. (2016). *Designing for life: A human perspective on technology development*. Palgrave Macmillan.
Saariluoma, P., & Jokinen, J. (2014). Emotional dimensions of user experience? A user psychological analysis. *International Journal of Human-Computer Interaction*, *30*(4), 303–320. https://doi.org/10.1080/10447318.2013.858460
Saariluoma, P., & Karvonen, A. (2024). Theory languages in designing artificial intelligence. *AI & Society*, *39*(5), 2249–2258. https://doi.org/10.1007/s00146-023-01716-y
Saariluoma, P., & Oulasvirta, A. (2010). User psychology: Re-assessing the boundaries of a discipline. *Psychology*, *01*(05), 317–328. https://doi.org/10.4236/psych.2010.15041
Saariluoma, P., & Rauterberg, M. (2016). Turing's error-revised. *International Journal of Philosophy Study*, *4*(0), 22–41. https://doi.org/10.14355/ijps.2016.04.004
Saariluoma, P., & Rousi, R., et al. (2020). Emotions and technoethics. In Rousi (Eds.), *Emotions in technology design: From experience to ethics*. Springer International Publishing AG. https://doi.org/10.1007/978-3-030-53483-7_11
Salo-Pöntinen, H. (2021). AI ethics: Critical reflections on embedding ethical frameworks in AI technology. In M. Rauterberg (Ed.), *Culture and computing: Design thinking and cultural computing. 9th International Conference (C&C '21), 23rd HCI International Conference (HCII '21), virtual event, July 24–29, 2021. Proceedings, Part II* (pp. 311–329). Springer. https://doi.org/10.1007/978-3-030-77431-8_20
Salo-Pöntinen, H., & Saariluoma, P. (2022). Reflections on the human role in AI policy formulations: How do national AI strategies view people? *Discover Artificial Intelligence*, *2*(1), 3. https://doi.org/10.1007/s44163-022-00019-3
Scherer, K. R. (2009). The dynamic architecture of emotion: Evidence for the component process model. *Cognition and Emotion*, *23*(7), 1307–1351. https://doi.org/10.1080/02699930902928969
Scherer, K. R. (1994). Toward a concept of "modal emotions". In P. Ekman & R. J. Davidson (Eds.), *The nature of emotion: Fundamental questions* (pp. 25–31). Oxford University Press.
Scherer, K. R., Dieckmann, A., Unfried, M., Ellgring, H., & Mortillaro, M. (2021). Investigating appraisal-driven facial expression and inference in emotion communication. *Emotion (Washington, D.C.)*, *21*(1), 73–95. https://doi.org/10.1037/emo0000693
Scherer, K. R., & Moors, A. (2019). The emotion process: Event appraisal and component differentiation. *Annual Review of Psychology*, *70*(1), 1, 719–745. https://doi.org/10.1146/annurev-psych-122216-011854
Scherer, K. R., Mortillaro, M., Rotondi, I., Sergi, I., & Trznadel, S. (2018). Appraisal-driven facial actions as building blocks for emotion inference. *Journal of Personality and Social Psychology*, *114*(3), 358–379. https://doi.org/10.1037/pspa0000107
Scherer, K. R., Mortillaro, M., & Mehu, M. (2017). Facial expression is driven by appraisal and generates appraisal inference. In J.-M. Fernández-Dols and J. A. Russell (Eds.), *The science of facial expression* (pp. 353–373). Oxford University Press.
Schwartz, S. H. (2012). An overview of the Schwartz theory of basic values. *Online Readings in Psychology and Culture*, *2*(1). https://doi.org/10.9707/2307-0919.1116
Seetha, J., Ayyadurai, M., & Florence, M. (2024). Novel hybrid optimal deep network and optimization approach for human face emotion recognition. In S. Ghosh, M. Niranjanamurthy, K. Deyasi, B. Basu Mallik, & S. Das (Eds.), *Mathematics and computer science* (Vol. 2, pp. 59–75). Wiley-Scrivener.
Shilton, K. (2018). Values and ethics in human-computer interaction. *Foundations and Trends® in Human–Computer Interaction*, *12*(2), 107–171. https://doi.org/10.1561/1100000073
Shilton, K. (2013). Values levers: Building ethics into design. *Science, Technology, & Human Values*, *38*(3), 374–397. https://doi.org/10.1177/0162243912436985
Shuman, V., Sander, D., & Scherer, K. R. (2013). Levels of valence. *Frontiers in Psychology*, *4*, 261. https://doi.org/10.3389/fpsyg.2013.00261
Siegel, E. H., Sands, M. K., Van den Noortgate, W., Condon, P., Chang, Y., Dy, J., Quigley, K. S., & Barrett, L. F. (2018). Emotion fingerprints or emotion populations? A meta-analytic investigation of autonomic features of emotion categories. *Psychological Bulletin*, *144*(4), 343–393. https://doi.org/10.1037/bul0000128
Sigfrids, A., Leikas, J., Salo-Pöntinen, H., & Koskimies, E. (2023). Human-centricity in AI governance: A systemic approach. *Frontiers in Artificial Intelligence*, *6*, 976887. https://doi.org/10.3389/frai.2023.976887
Simon, H. A. (1970). *The sciences of the artificial*. The MIT Press.

Siponen, M., & Klaavuniemi, T. (2021). Demystifying beliefs about the natural sciences in information system. *Journal of Information Technology*, *36*(1), 56–68. https://doi.org/10.1177/0268396220901535
Skinner, B. F. (1974). *About behaviorism*. Vintage.
Skinner, B. F. (1977). Why I am not a cognitive psychologist. *Behaviorism*, *5*(2), 1–10. http://www.jstor.org/stable/27758892
Sloane, M., Moss, E., & Chowdhury, R. (2022). A Silicon Valley love triangle: Hiring algorithms, pseudo-science, and the quest for auditability. *Patterns (New York, N.Y.)*, *3*(2), 100425. https://doi.org/10.1016/j.patter.2021.100425
Søndergaard, M. L. J. (2025). What Mosses can teach us about design fabulations and feminist more-than-human care. *Human–Computer Interaction*, *40*(1–4), 43–64. https://doi.org/10.1080/07370024.2023.2269893
Stark, L. (2023). Artificial intelligence and the conjectural sciences. *BJHS Themes*, *8*, 35–49. https://doi.org/10.1017/bjt.2023.3
Stark, L., & Hoey, J. (2021). The ethics of emotion in artificial intelligence systems. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)* (pp. 782–793). Association for Computing Machinery. https://doi.org/10.1145/3442188.3445939
Stark, L., & Hutson, J. (2022). Physiognomic artificial intelligence. *Fordham Intellectual Property, Media & Entertainment Law Journal*, *32*(4), 922–978. https://ir.lawnet.fordham.edu/iplj/vol32/iss4/2
Stephanidis, C., Salvendy, G., Antona, M., Duffy, V. G., Gao, Q., Karwowski, W., Konomi, S., Nah, F., Ntoa, S., Rau, P. P., Siau, K., & Zhou, J. (2025). Seven HCI grand challenges revisited: Five-year progress. *International Journal of Human–Computer Interaction*, 1–49. https://doi.org/10.1080/10447318.2025.2450411
Sultana, S., & Ahmed, S. I. (2019). Witchcraft and HCI: Morality, modernity, and postcolonial computing in rural Bangladesh. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)* (Paper 356, pp. 1–15). Association for Computing Machinery. https://doi.org/10.1145/3290605.3300586
Tangney, J. P., Stuewig, J., & Mashek, D. J. (2007). Moral emotions and moral behavior. *Annual Review of Psychology*, *58*(1), 345–372. https://doi.org/10.1146/annurev.psych.56.091103.070145
Thagard, P. (2023). Cognitive Science. In: Edward N. Zalta & Uri Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2023 ed.). https://plato.stanford.edu/archives/win2023/entries/cognitive-science/
Todorov, A. (2017). *Face value: The irresistible influence of first impressions*. Princeton University Press.
Todorov, A., Olivola, C. Y., Dotsch, R., & Mende-Siedlecki, P. (2015). Social attributions from faces: Determinants, consequences, accuracy, and functional significance. *Annual Review of Psychology*, *66*(1), 1, 519–545. https://doi.org/10.1146/annurev-psych-113011-143831
Tzeng, S. (2022). Engineering normative and cognitive agents with emotions and values. In *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems (AAMAS '22)* (pp. 1878–1880). International Foundation for Autonomous Agents and Multiagent Systems. https://doi.org/10.5555/3535850.3536137
Uddin, M. T., Yin, L., & Canavan, S. (2024). Spatio-temporal graph analytics on secondary affect data for improving trustworthy emotional AI. *IEEE Transactions on Affective Computing*, *15*(1), 30–49. https://doi.org/10.1109/TAFFC.2023.3296695
Ullstein, C., Engelmann, S., Papakyriakopoulos, O., Hohendanner, M., & Grossklags, J. (2022). AI-competent individuals and laypeople tend to oppose facial analysis AI. In *Proceedings of Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '22)* (Vol. 9, pp. 1–12). Association for Computing Machinery. https://doi.org/10.1145/3551624.3555294
Varanka, T., Li, Y., Peng, W., & Zhao, G. (2024). Data leakage and evaluation issues in micro-expression analysis. *IEEE Transactions on Affective Computing*, *15*(1), 186–197. https://doi.org/10.1109/TAFFC.2023.3265063
Vianello, A., Laine, S., & Tuomi, E. (2023). Improving trustworthiness of AI solutions: A qualitative approach to support ethically-grounded AI design. *International Journal of Human–Computer Interaction*, *39*(7), 1405–1422. https://doi.org/10.1080/10447318.2022.2095478
von Wright, G. H. (1971). *Explanation and understanding*. Routledge.
Wachter, S., Mittelstadt, B., & Russell, C. (2021). Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Computer Law & Security Review*, *41*, 105567. ISSN 0267-3649. https://doi.org/10.1016/j.clsr.2021.105567
Wallach, W., Franklin, S., & Allen, C. (2010). A conceptual and computational model of moral decision making in human and artificial agents. *Topics in Cognitive Science*, *2*(3), 454–485. https://doi.org/10.1111/j.1756-8765.2010.01095.x
Wallach, W., & Allen, C. (2009). *Moral machines: Teaching robots right from wrong*. Oxford University Press.
Wei, J., Hu, G., Yang, X., Luu, A. T., & Dong, Y. (2024). Learning facial expression and body gesture visual information for video emotion recognition. *Expert Systems with Applications*, *237*(Part A), 121419. https://doi.org/10.1016/j.eswa.2023.121419
Westermarck, E. (1906). *The origins and development of moral ideas*. MacMillan.
Wittgenstein, L. (1953). *Philosophical investigations: Philosophische Untersuchungen.* (G. E. M. Anscombe, ed. & trans.). Macmillan.
Wu, L., Winkler, M. H., Andreatta, M., Hajcak, G., & Pauli, P. (2012). Appraisal frames of pleasant and unpleasant pictures alter emotional responses as reflected in self-report and facial electromyographic activity. *International*

Journal of Psychophysiology: Official Journal of the International Organization of Psychophysiology, *85*(2), 224–229. https://doi.org/10.1016/j.ijpsycho.2012.04.010

Yang, D., Kommineni, A., Alshehri, M., Mohanty, N., Modi, V., Gratch, J., & Narayanan, S. (2023). *Context unlocks emotions: Text-based emotion classification dataset auditing with large language models* [Paper presentation]. 11th International Conference on Affective Computing and Intelligent Interaction (ACII), 2023, Cambridge, MA, pp. 1–8. https://doi.org/10.1109/ACII59096.2023.10388131

Yang, J., Zhou, K., Li, Y., & Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, *132*(12), 5635–5662. https://doi.org/10.1007/s11263-024-02117-4

Yin, R. K. (2009). *Case study research. Design and methods* (Vol. 5.). SAGE.

Zajonc, R. (2000). Feeling and thinking: Closing the debate over the independence of affect. In Joseph P. Forgas (Ed.), *Feeling and thinking: The role of affect in social cognition* (pp. 31–58). Cambridge University Press.

Zhang, L., & Arandjelović, O. (2021). Review of automatic microexpression recognition in the past decade. *Machine Learning and Knowledge Extraction*, *3*(2), 414–434. https://doi.org/10.3390/make3020021

Zhang, L., Wang, S., & Liu, B. (2018). Deep learning for sentiment analysis: A survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, *8*(4), e1253. https://doi.org/10.1002/widm.1253

Zhang, M., Marklund, H., Dhawan, N., Gupta, A., Levine, S., & Finn, C. (2021). Adaptive Risk Minimization: Learning to Adapt to Domain Shift. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS 2021)* (Vol 34, pp. 23664–23678).

Zhang, S., Yang, Y., Chen, C., Zhang, X., Leng, Q., & Zhao, X. (2024). Deep learning-based multimodal emotion recognition from audio, visual, and text modalities: A systematic review of recent advancements and future prospects. *Expert Systems with Applications*, *237*(Part C), 121692. https://doi.org/10.1016/j.eswa.2023.121692

Zhang, J., Zheng, K., Mazhar, S., Fu, X., & Kong, J. (2023). Trusted emotion recognition based on multiple signals captured from video. *Expert Systems with Applications*, *233*, 120948. https://doi.org/10.1016/j.eswa.2023.120948

Zhu, J., & Thagard, P. (2002). Emotion and action. *Philosophical Psychology*, *15*(1), 19–36. https://doi.org/10.1080/09515080120109397

Zicari, R. V., Brodersen, J., Brusseau, J., Dudder, B., Eichhorn, T., Ivanov, T., Kararigas, G., Kringen, P., McCullough, M., Moslein, F., Mushtaq, N., Roig, G., Sturtz, N., Tolle, K., Tithi, J. J., van Halem, I., & Westerlund, M. (2021). Z-Inspection®: A process to assess trustworthy AI. *IEEE Transactions on Technology and Society*, *2*(2), 83–97. https://doi.org/10.1109/TTS.2021.3066209

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